Lecture 0:
인공지능수학 개요

Sung In Cho

Dept. of Artificial Intelligence
Sogang University

Introduction

- Mathematics for Artificial Intelligence (AIE3090)

- Lecturer

1. Cho, Sung In
    - Office: CY 407
    - Phone: -
    - E-mail: csi2267@sogang.ac.kr
    - Homepage: https://sites.google.com/view/csi2267svm

- Grade

1. Attendance (10%)
2. Midterm Exams (35%)
3. Final Exams (35%)
4. Homework (20%)

Introduction

- Prerequisites—*these are essential*!

1. Linear algebra
2. Vector calculus

Introduction

- Contact

1. E-mail (cell-phone ×)

- Illegal attendance or cheating in a test

1. -3 grades

- During the class (offline lecture)

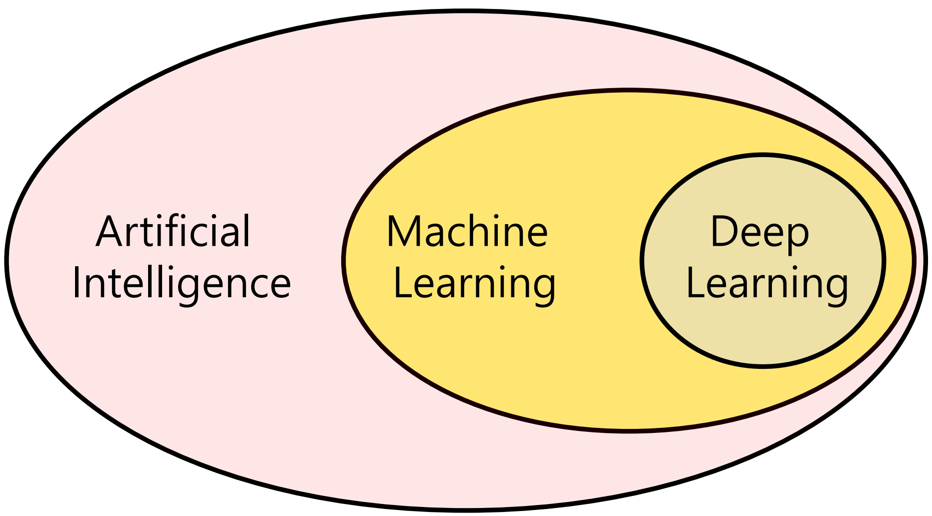
1. Cell-phone usage→ OUT (considered absent)

Introduction to AI

- Artificial intelligence

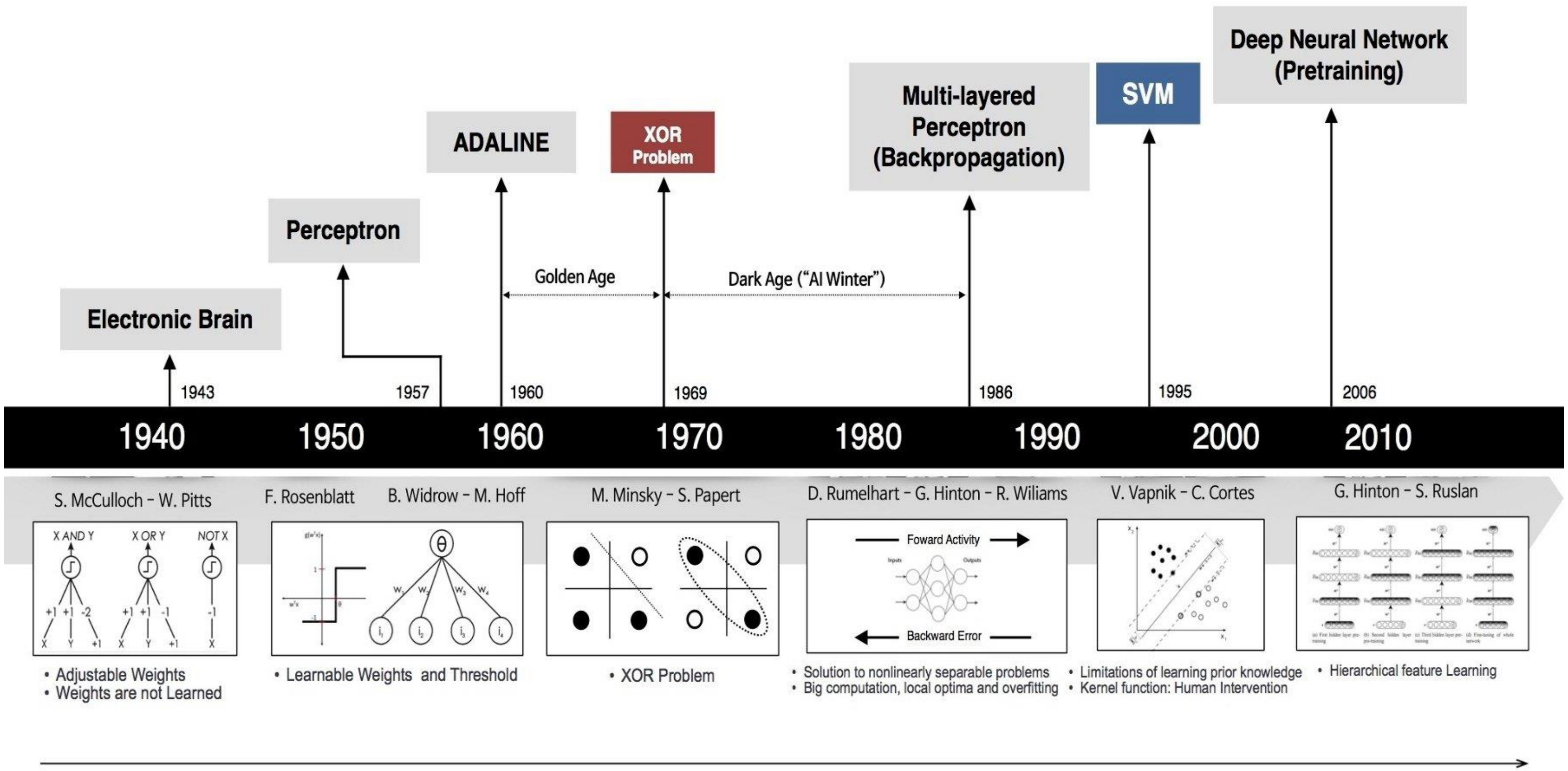
1. A field that artificially implements human learning, reasoning, and perception abilities

2. Artificial Intelligence > Machine Learning > Deep Learning
   - AI: Computers that mimic human thinking
   - Machine Learning: Computers that learn on their own
   - Deep Learning (Artificial Neural Network): A technology that implements machine learning, a learning method inspired by human brain



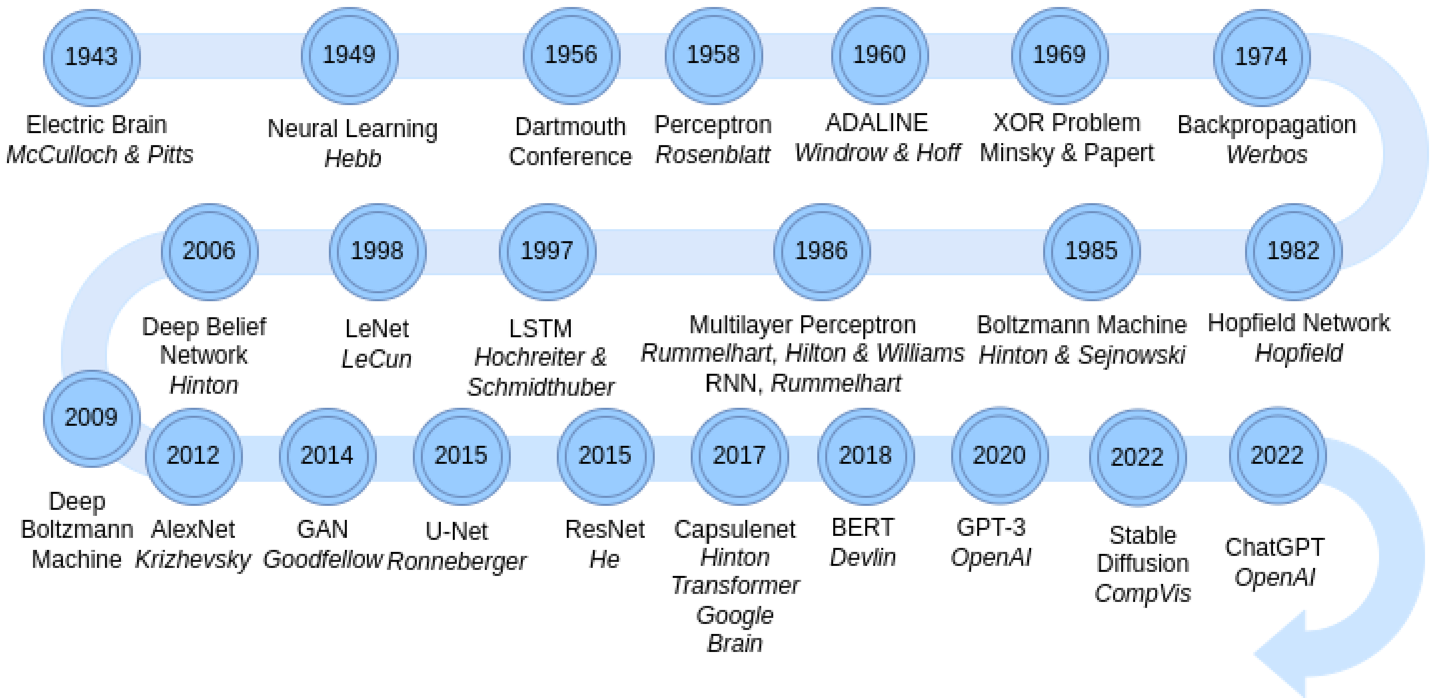
Smart Vision & Media Lab.

- A brief history of neural nets (Early era)



Sogang University

- A brief history of neural nets

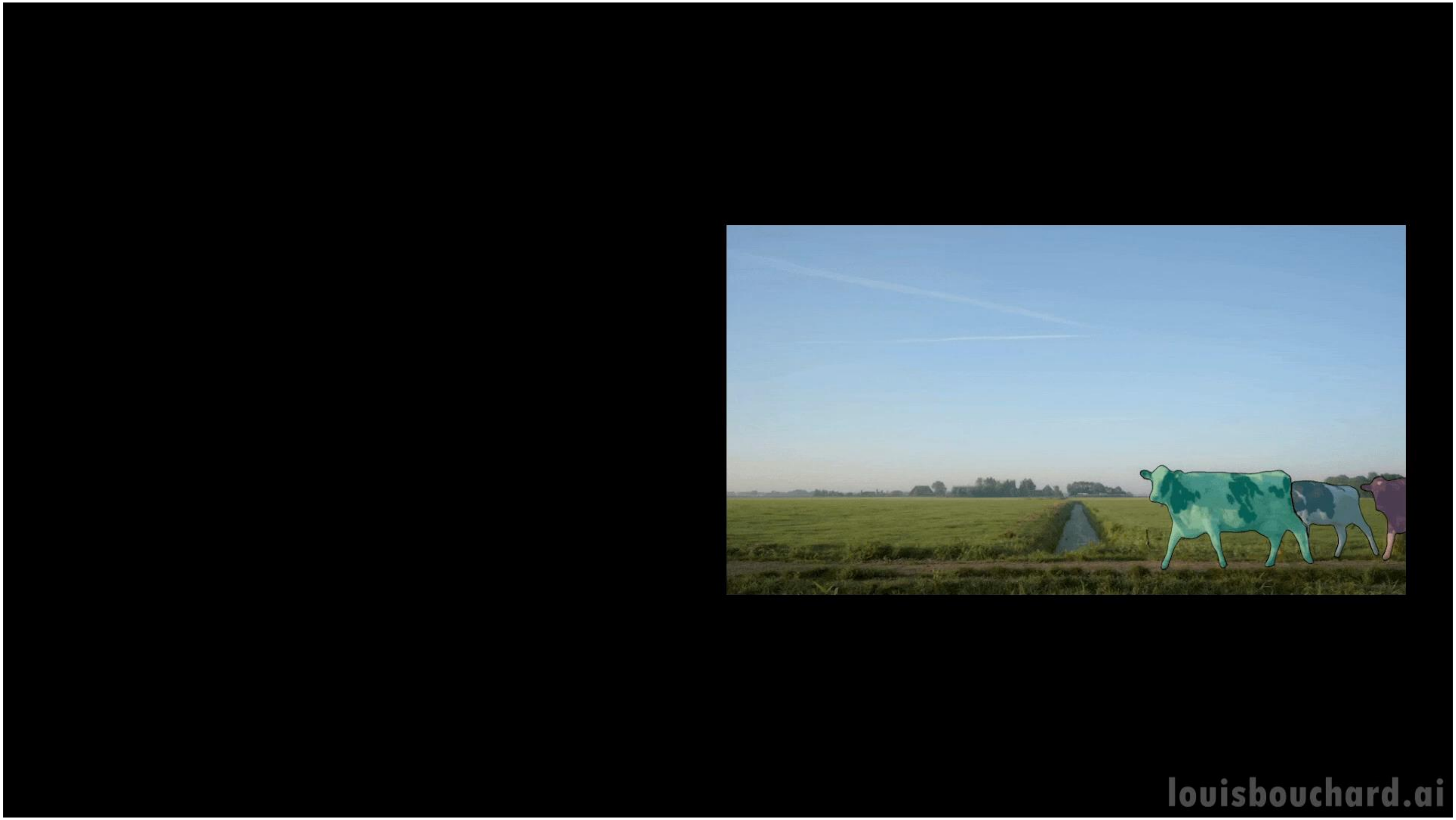


Applications of Artificial Intelligence

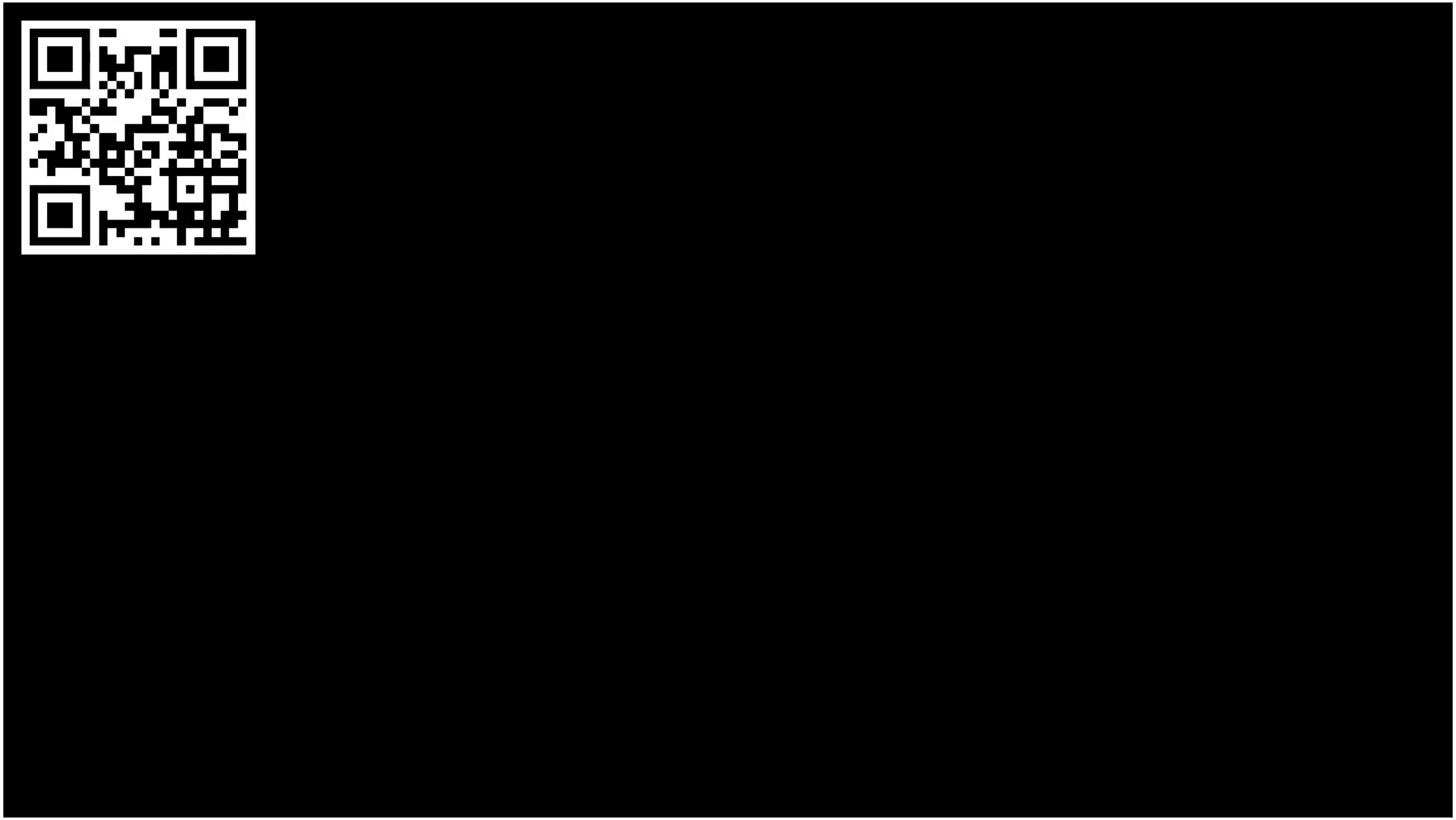
- Drive Sim

- Ultra-Fast-Lane-Detection-V2

- Segment Anything



- Generative Model



Applications of Artificial Intelligence

- DALL-E3

A brain riding a rocketship heading towards the moon

A Toronto skyline with Google brain Logo written in fireworks

Close-up photograph of a hermit crab nestled in wet sand, with sea foam nearby and the details of its shell and texture of the sand accentuated

- Imagen

A brain riding a rocketship heading towards the moon

A Toronto skyline with Google brain Logo written in fireworks

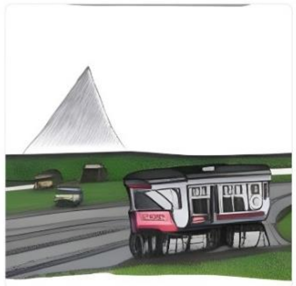
A giant cobra snake on a farm. The snake is made out of corn.

A dog looking curiously in the mirror, seeing a cat

Applications of Artificial Intelligence

▪ Make-a-scene

A drawing of a train



A monster robot bear riding a train

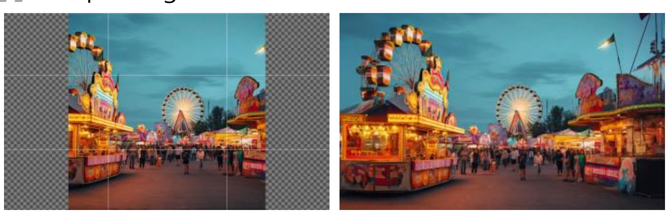
▪ Generative-Fil

" Text-to-Image "

Friendly dragon emerging from stormy sea

" Inpainting "

" Outpainting "



" Style transfer "

Purple gloss balloon

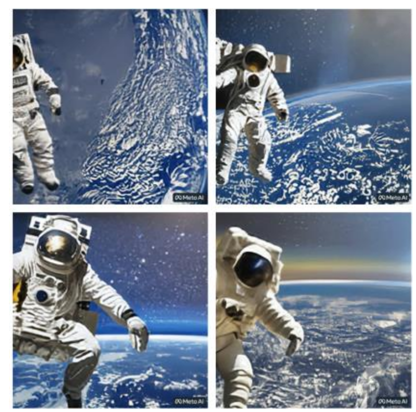
- Make-a-video

" Text-to-Video "

A teddy bear painting a portrait

" Single image-to-Video "

" Create variations "



Applications of Artificial Intelligence

- Sora

A stylish woman walks down a Tokyo street filled with warm glowing neon and animated city signage. She wears a black leather jacket, a long red dress, and black boots, and carries a black purse. She wears sunglasses and red lipstick. She walks confidently and casually. The street is damp and reflective, creating a mirror effect of the colorful lights. Many pedestrians walk about.

- Sora

Historical footage of California during the gold rush.

Applications of Artificial Intelligence

- Kling

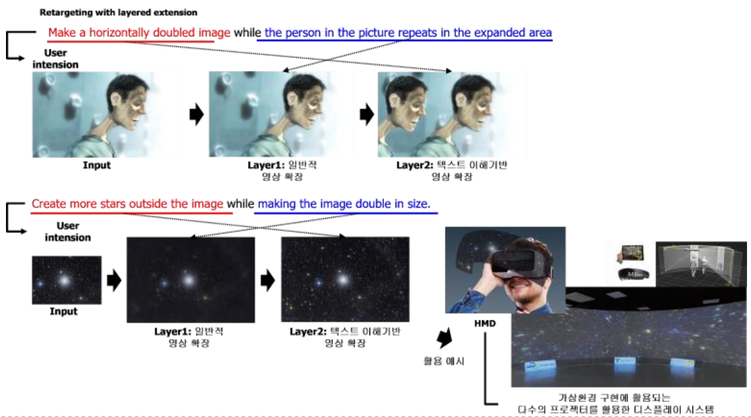
A small white rabbit wearing glasses sits on a chair in a café reading a newspaper, with a cup of hot coffee on the table

2023.03.01 ~ 2026.02.28 (On-going)

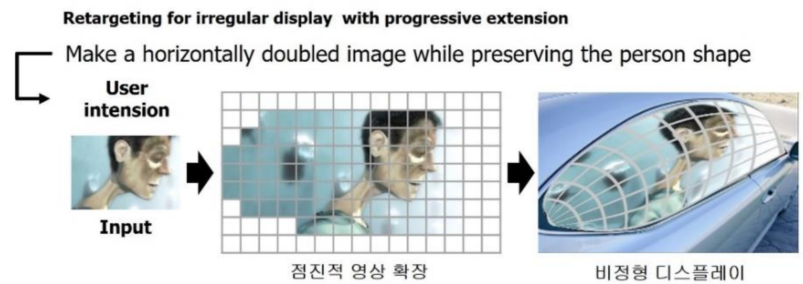
▪ Component technology for retargeting based on video content expansion

1. Enhancing object-centric image expansion via Large Language Model (LLM)-based video object information extraction and diffusion-based inpainting

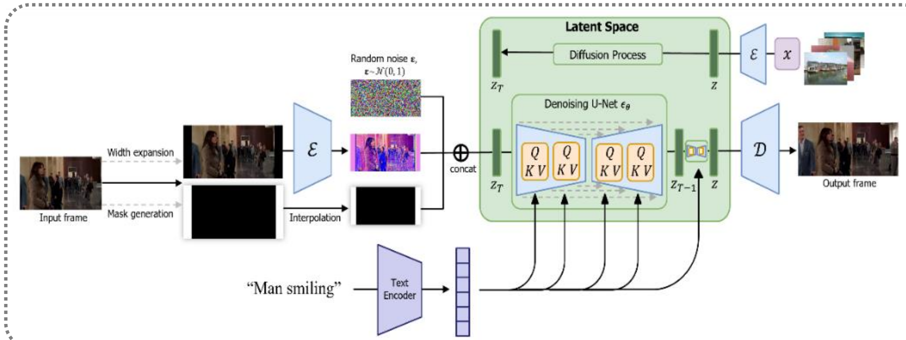
<Research objective>



<Progressive video expansion for non-rectangular displays>



<Stable diffusion model-based text-driven video expansion architecture>



<Stable diffusion model-based video expansion example>

Input output Input output

Smart Vision & Media Lab.

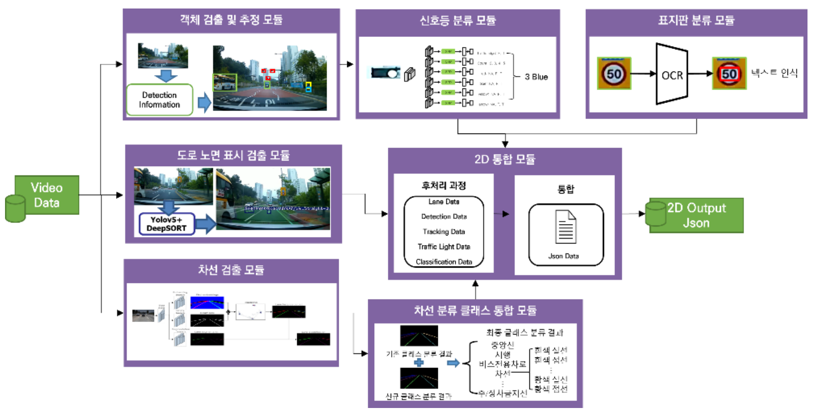
2021.04.01 ~ 2024.12.31 (Completed)

경찰청

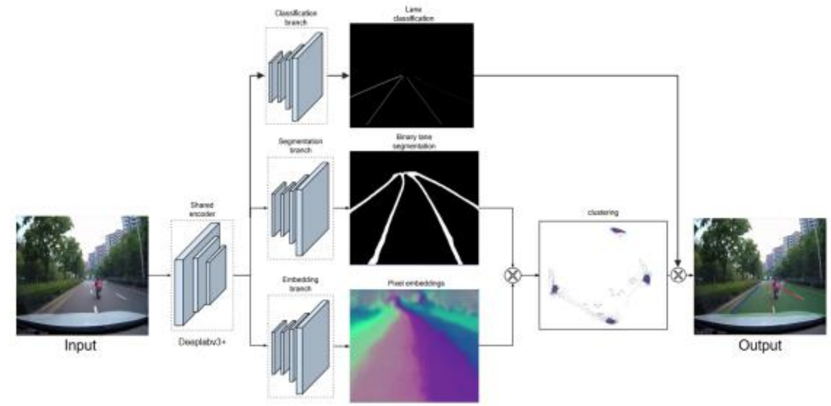
- Standardizing AI driving ability assessment and developing the evaluation process

1. Deep learning-based forward perception system development

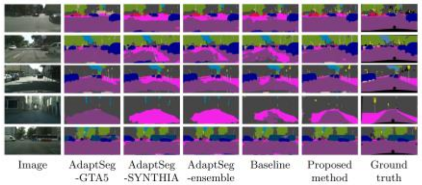
<System overview diagram>



<Lane detection model structure>



<Detection and classification visualization results>



Road region segmentation using domain adaptation techniques

Static and dynamic object detection and classification results

<Demo video>

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Applications of Artificial Intelligence

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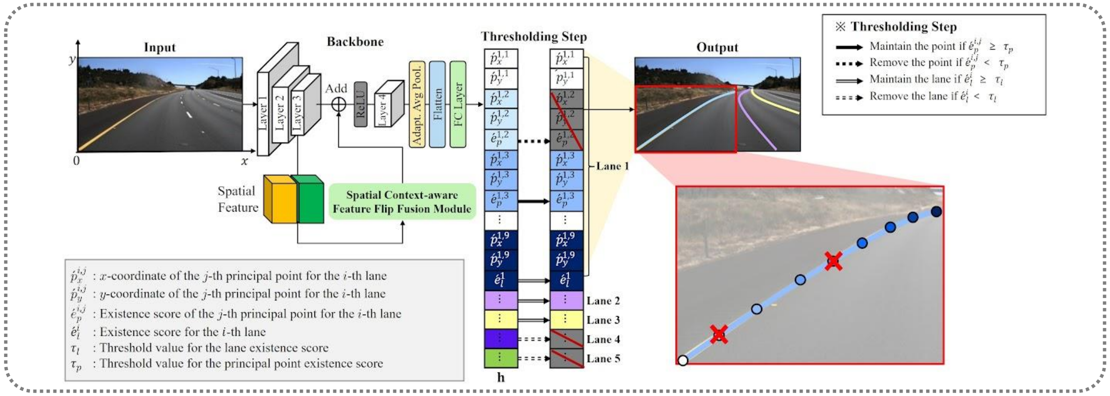
2021.04.01 ~ 2024.12.31 (Completed)

경찰청

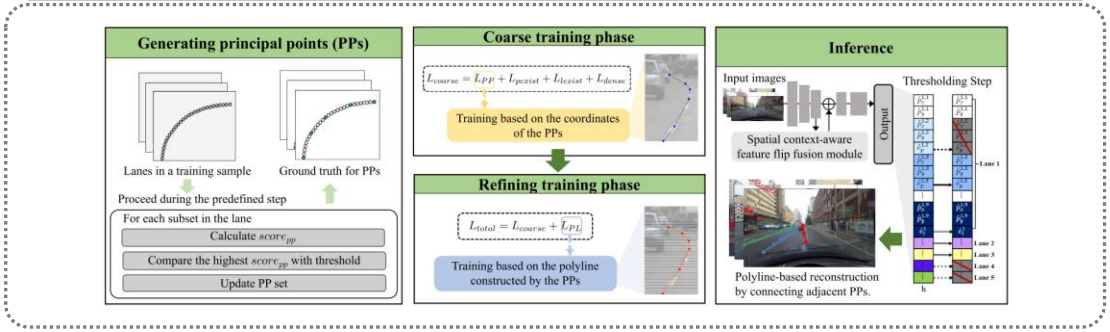
- Lane Detection (IT-ITS 2023, [JCR 2.5%, IF: 7.9])

1. Polyline formation based on key-point coordinates for lane detection and representation
2. Output an optimized set of points for each lane in road images without post-processing

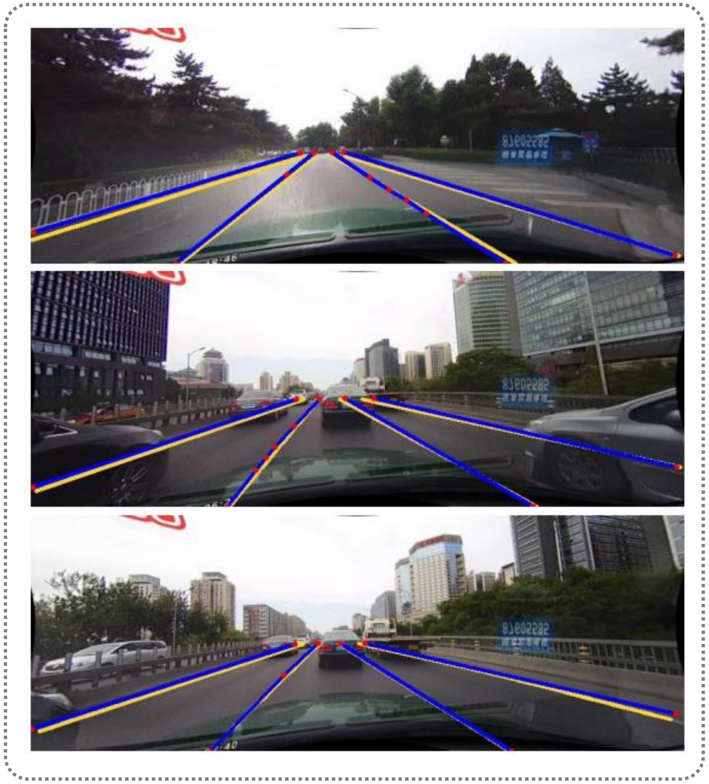
<Overall flowchart of the proposed method>



<Algorithm flowchart of the proposed method>



<Lane detection results>

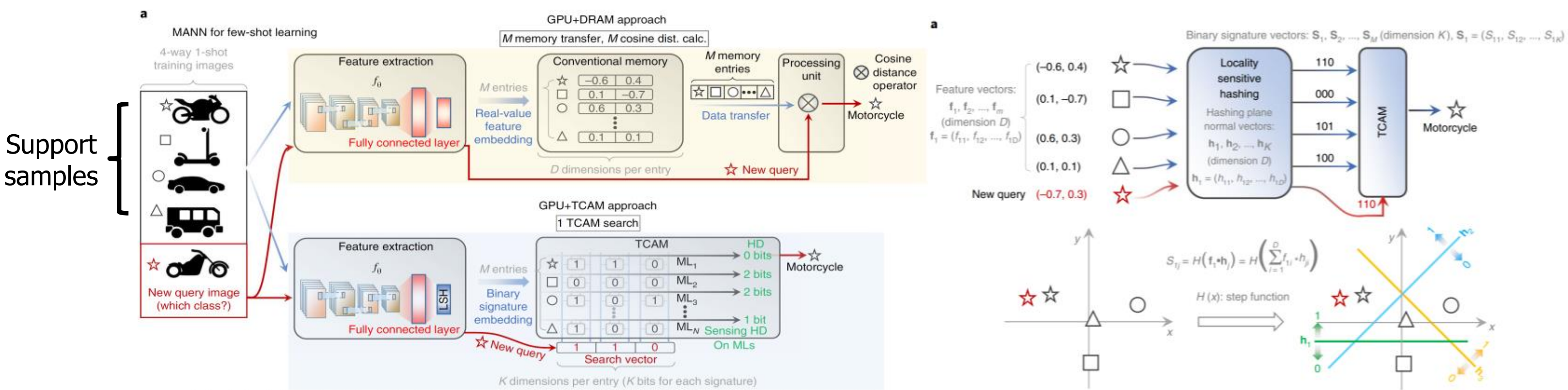


*Y. J. Chae, S. J. Park, E. S. Kang, M. J. Chae, B. H. Ngo, and S. I. Cho, "Point2Lane: Polyline-based Reconstruction with Principal Points for Lane Detection," in IEEE Transactions on Intelligent Transportation Systems

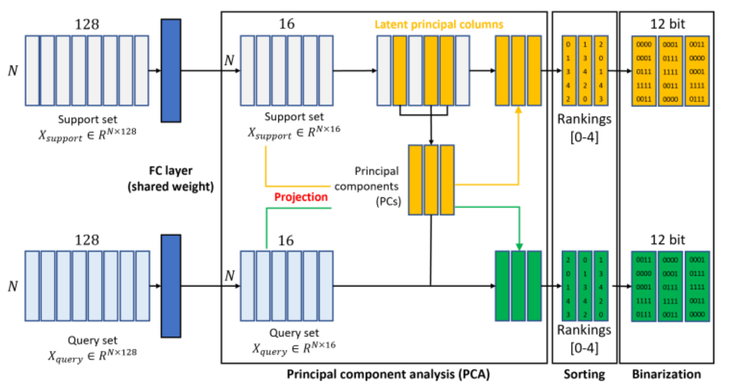
2022.04.15 ~ 2024.12.31 (Completed)

- Development of Locality Sensitive Hashing (LSH) for TCAM-based few-shot network implementation

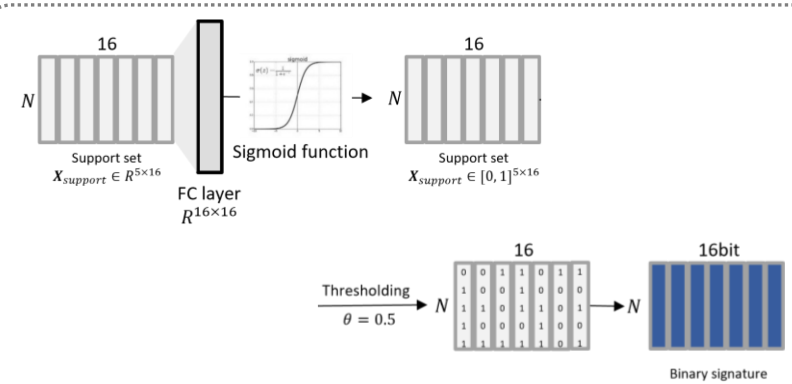
<Locality Sensitive Hashing (LSH) Function based Meta Learning>



<PCA based LSH Function generation enhancement>



<Sigmoid based Locality Sensitive Hashing (LSH) Function>



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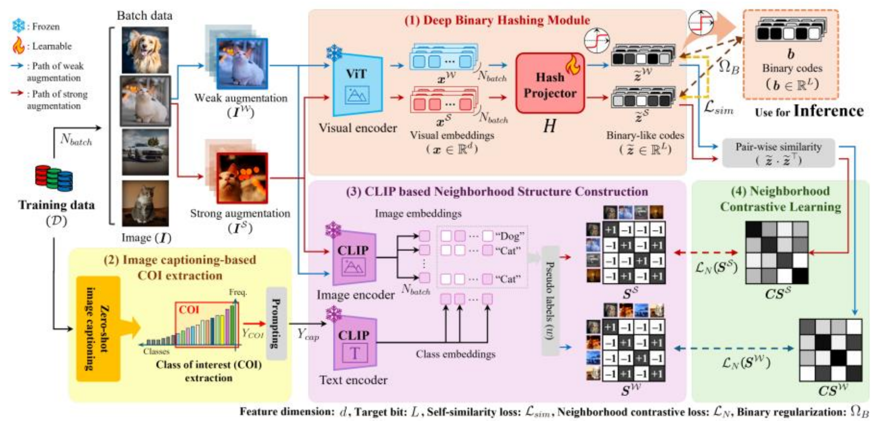
2022.04.15 ~ 2024.12.31 (Completed)

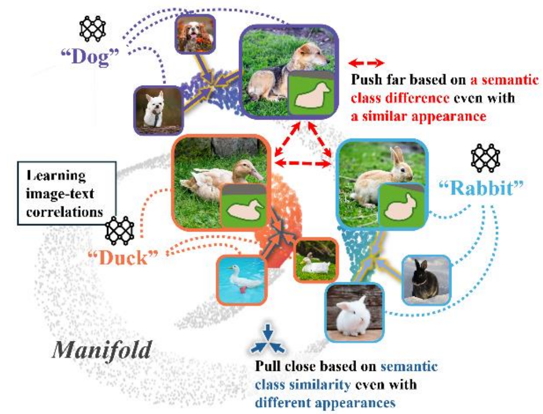
NRF 한국연구재단

- Development of Locality Sensitive Hashing (LSH) for TCAM-based few-shot network implementation

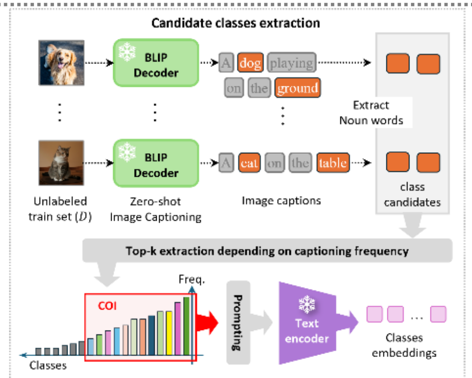
<Unsupervised Deep Binary Hashing for Image Retrieval*>



<Motivation concept diagram>



<Candidate class extraction process>



<Experimental results by dataset>

Method		CIFAR-10 (I)			CIFAR-10 (II)			NUS-WIDE			FLICKR25K		
		16 bits	32 bits	64 bits	16 bits	32 bits	64 bits	16 bits	32 bits	64 bits	16 bits	32 bits	64 bits
VGG+LSH	STOC'02	14.38	15.86	18.09	12.55	13.76	15.07	38.52	41.43	43.89	56.11	57.08	59.26
VGG+SH	IJAR'09	28.28	28.86	28.51	24.68	25.34	27.16	46.85	53.63	56.28	60.02	63.30	64.17
VGG+ITQ	TPAMI'12	34.41	35.41	38.82	30.50	32.50	34.90	62.70	64.50	66.40	63.30	65.92	68.86
DeepBit	CVPR'16	19.43	24.86	27.73	20.60	28.23	31.30	39.20	40.30	42.90	62.04	66.54	68.34
SGH	ICML'17	34.51	37.04	38.93	43.50	43.70	43.30	59.30	59.00	60.70	72.10	72.84	72.83
BGAN	AAAI'18	-	-	-	52.50	53.10	56.20	68.40	71.40	73.00	-	-	-
GreedyHash	NeurIPS'18	44.80	47.20	50.10	45.76	48.26	53.34	63.30	69.10	73.10	69.91	70.85	73.03
HashGAN	CVPR'18	44.70	46.30	48.10	42.81	47.54	47.29	68.44	70.56	71.71	72.11	73.25	75.46
TBH	CVPR'20	54.68	58.63	62.47	53.20	57.30	57.80	71.70	72.50	73.50	74.38	76.14	77.87
CIBHash	IJCAI'21	59.39	63.67	65.16	59.00	62.20	64.10	79.00	80.70	**81.50**	77.21	78.43	79.50
MeCoQ	AAAI'22	**68.20**	**69.74**	**71.06**	**62.88**	**64.09**	**65.07**	**80.18**	**82.16**	**83.24**	**81.31**	**81.71**	**82.68**
VLBH (ours)	-	**85.42**	**86.84**	**86.20**	**81.15**	**79.75**	**81.27**	**80.43**	**81.49**	81.28	**80.08**	**80.26**	**80.06**

*G. Lee, S. Park, S. Ahn, J. Y. Lee, J. H. Park, and S. I. Cho, "Vision-Language Pre-trained Models Define a New Era of Unsupervised Deep Binary Hashing for Image Retrieval" (Submitted).

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Applications of Artificial Intelligence

Smart Vision & Media Lab.

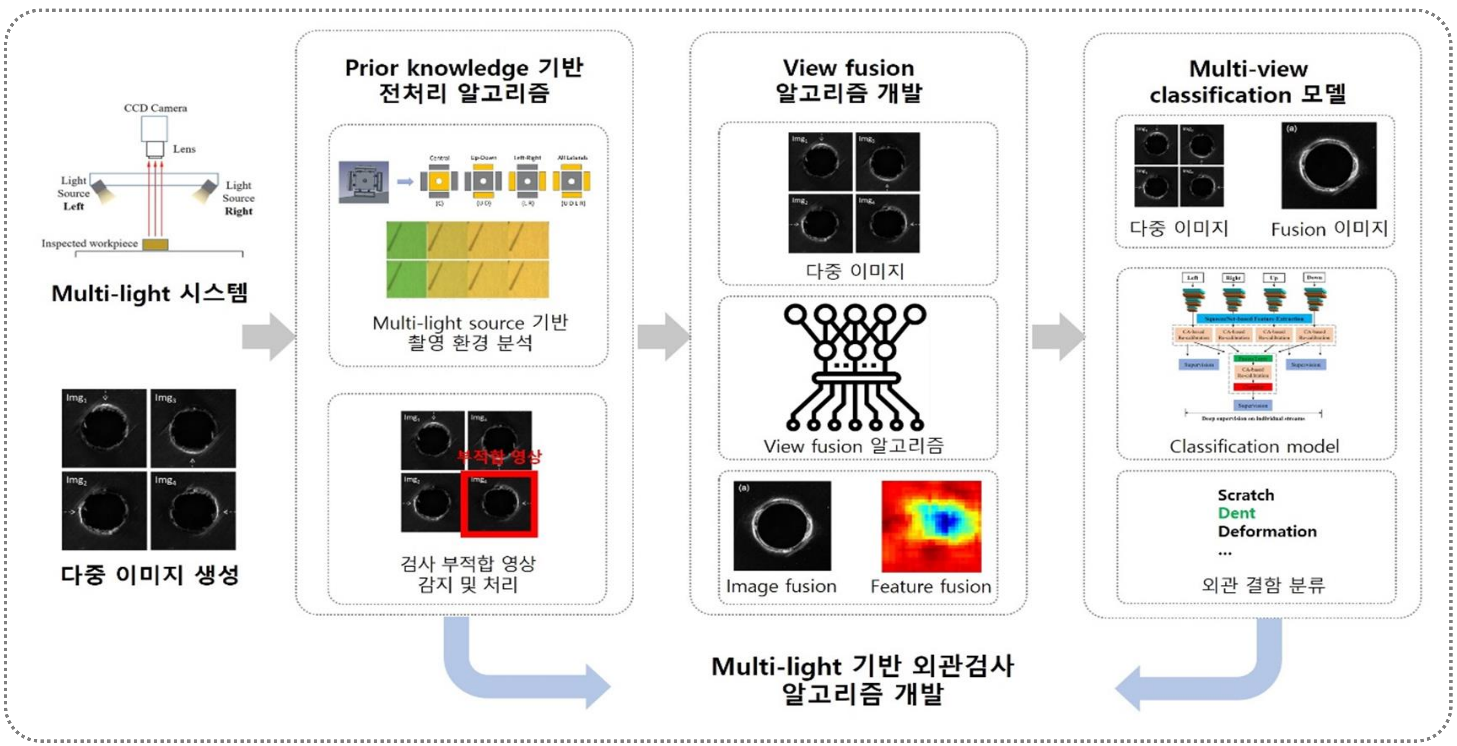
2024.03.01 ~ 2025.03.01 (Completed)

LG디스플레이

- Classification using multi-light-based image synthesis for product surface defect inspection

1. Preprocessing algorithm development based on prior knowledge
2. View fusion algorithm development
3. Multi-view classification model development

<Multi-light-based visual inspection algorithm>



Sogang University

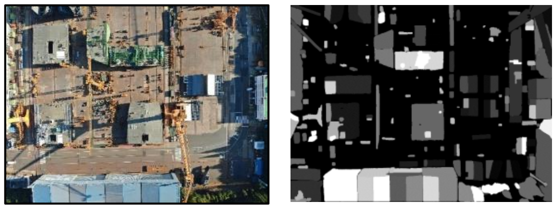
2023.03.01 ~ 2024.02.29 (Completed)

한화오션

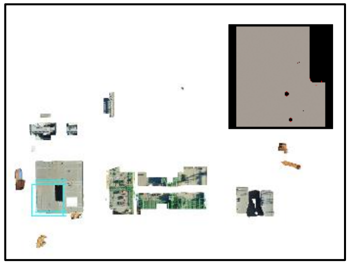
- Block process status assessment using textual schedule information and drone video data

<Change region extraction>

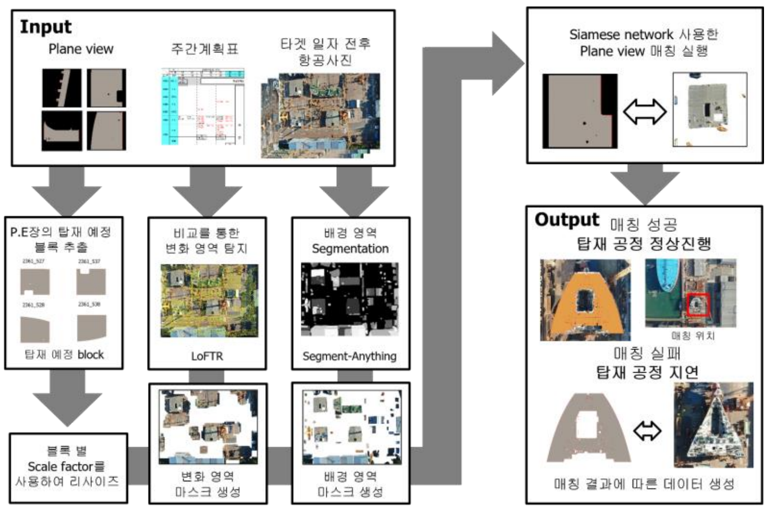
<Background segmentation with SAM>



<Scheduled block exploration>



<Mounting process monitoring>

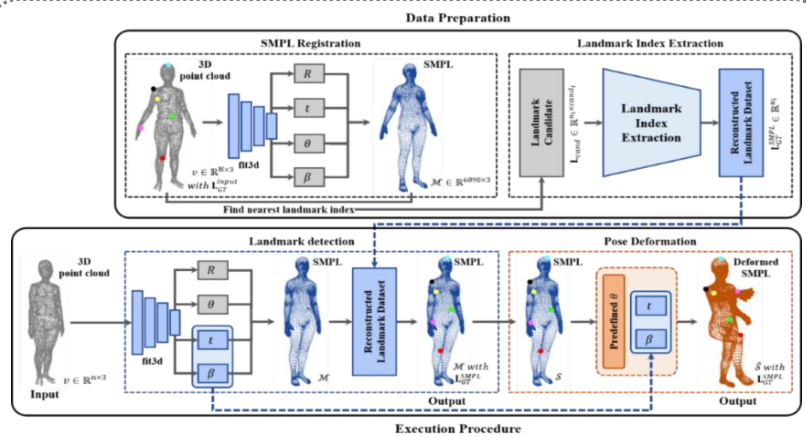


<Demo video>

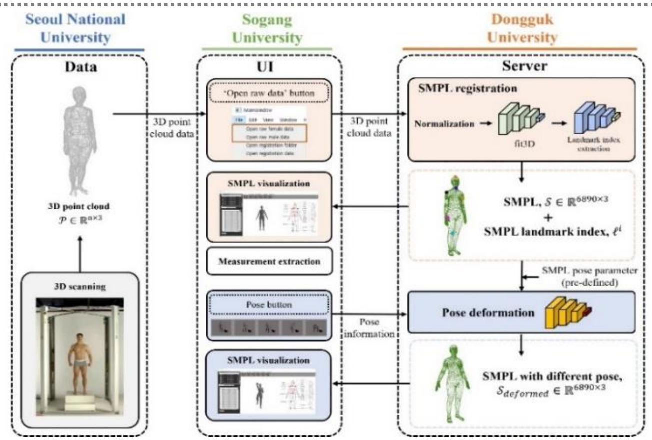
2023.06.01 ~ 2023.12.15 (Completed)

- Technology and procedure development for deep learning-based 3D dynamic n body surface shape analysis

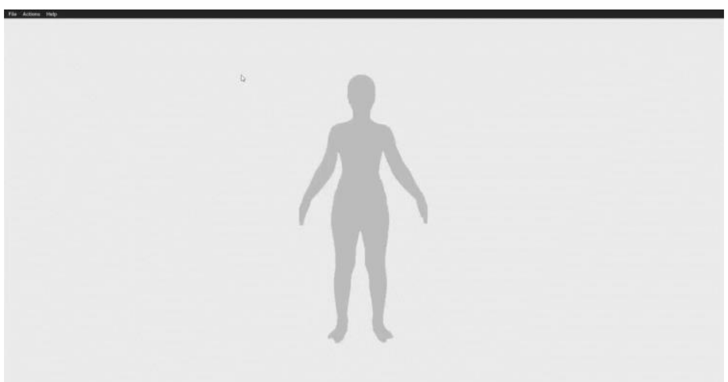
<Dynamic human body surface shape analysis technology operation structure>



<Server communication diagram>

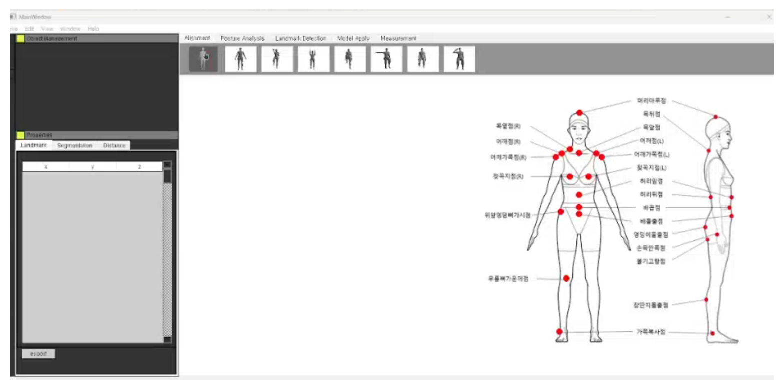


<Visualization results of detected landmarks>



: Ground-truth : SMPL landmark index

<Final 3D dynamic human body analysis program demo video>



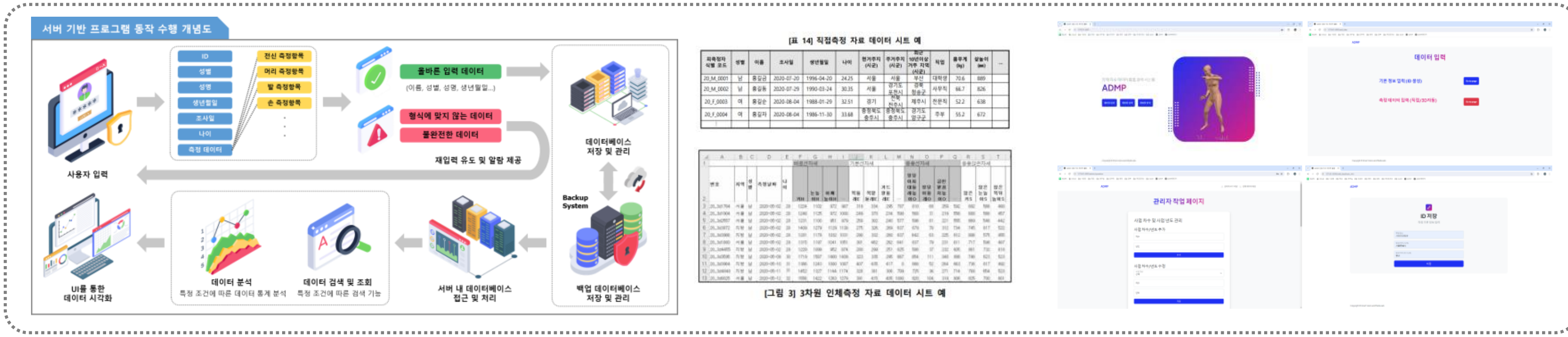
2024.06.01 ~ 2024.12.15 (Completed)

국가기술표준원

Katri Testing & Research Institute

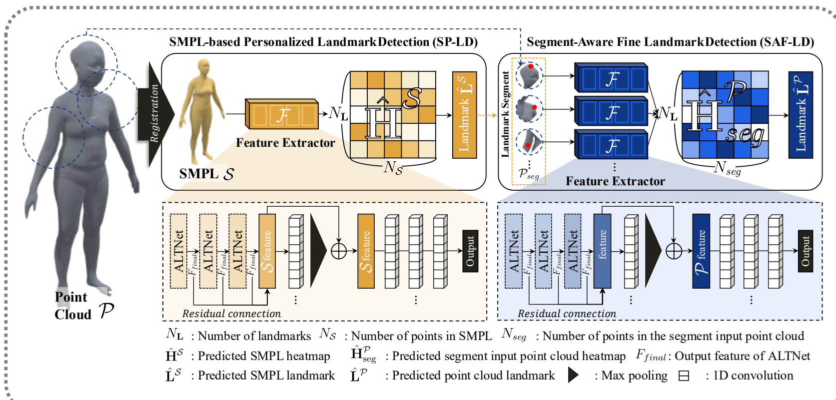
▪ Integrated management system for anthropometric data

<Integrated anthropometric data management system and database>

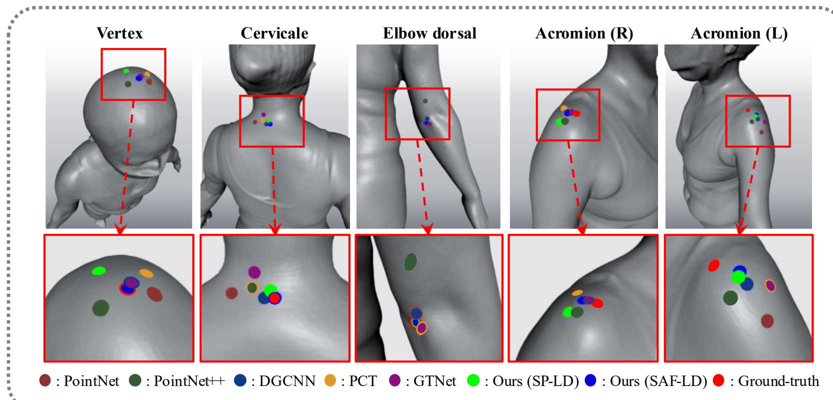


▪ Development of a deep learning-based method for 3D anthropometric landmark ext raction

<Process of the proposed method>



<Visualization results of detected landmarks>



*J. S. Byun, J. H. Park, M. H. Cha, J. Y. Kim, J. W. Park, S. I. Cho, "You Don't Need Dynamic Scan for 3D Anthropometric Landmark Detection," (Submitted).

Sogang University

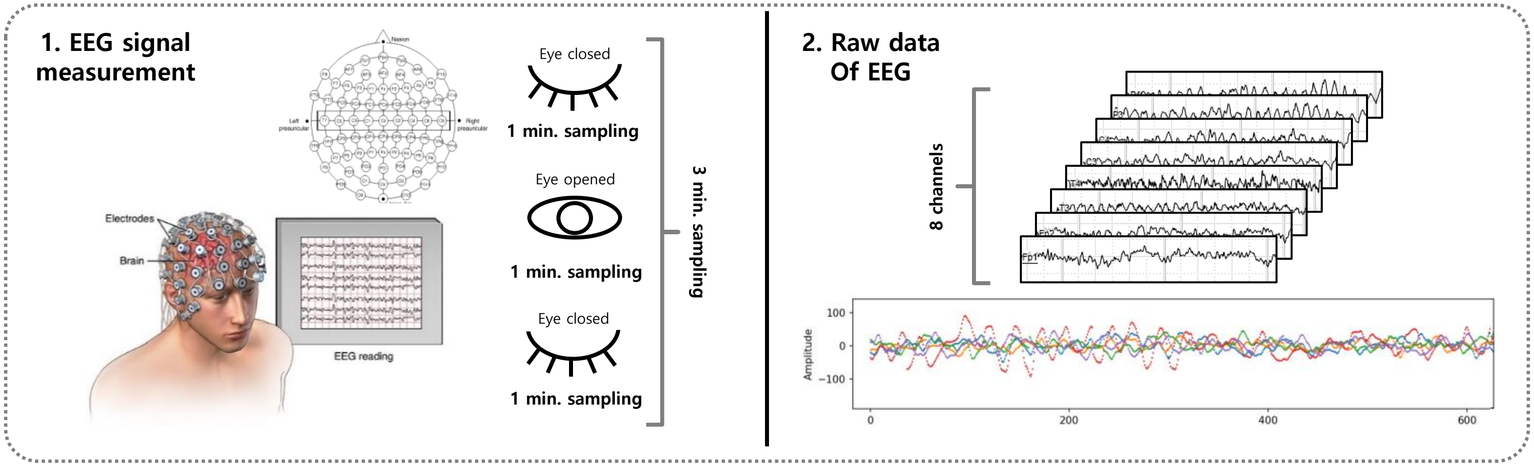
Applications of Artificial Intelligence

Smart Vision & Media Lab.

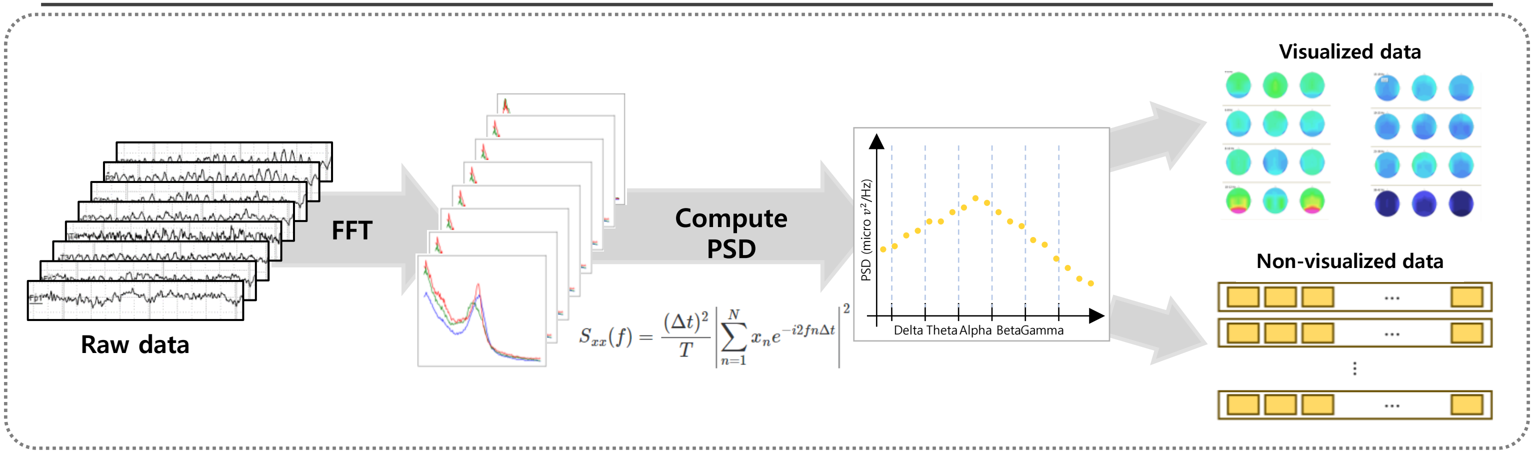
2024.06.01 ~ 2024.11.30 (Completed)

- ADHD detection using EEG signals with a deep learning model

<EEG signal acquisition process>



<EEG signal preprocessing process>



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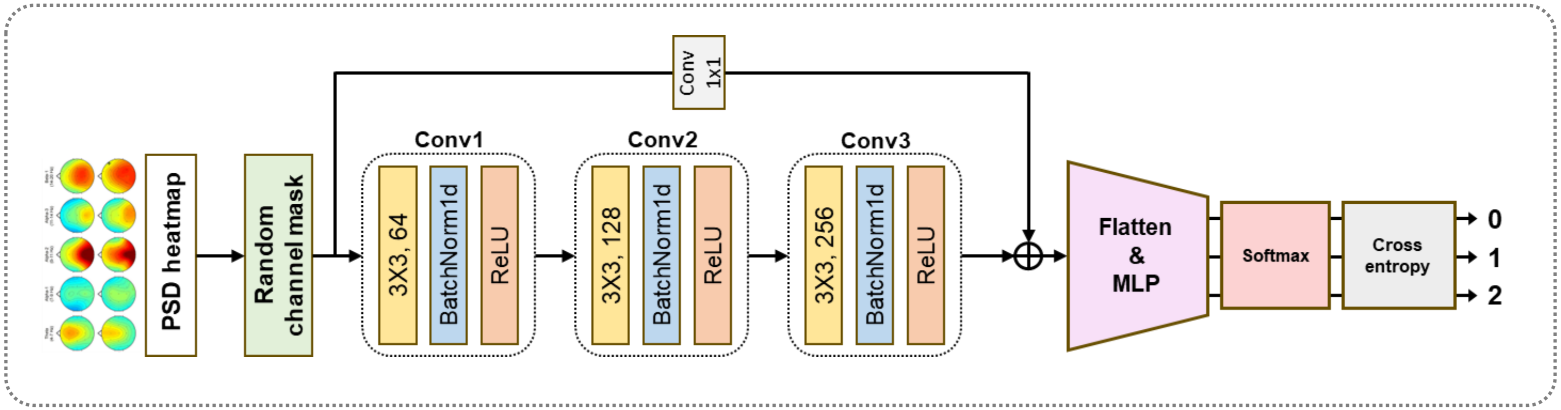
Smart Vision & Media Lab.

2024.06.01 ~ 2024.11.30 (Completed)

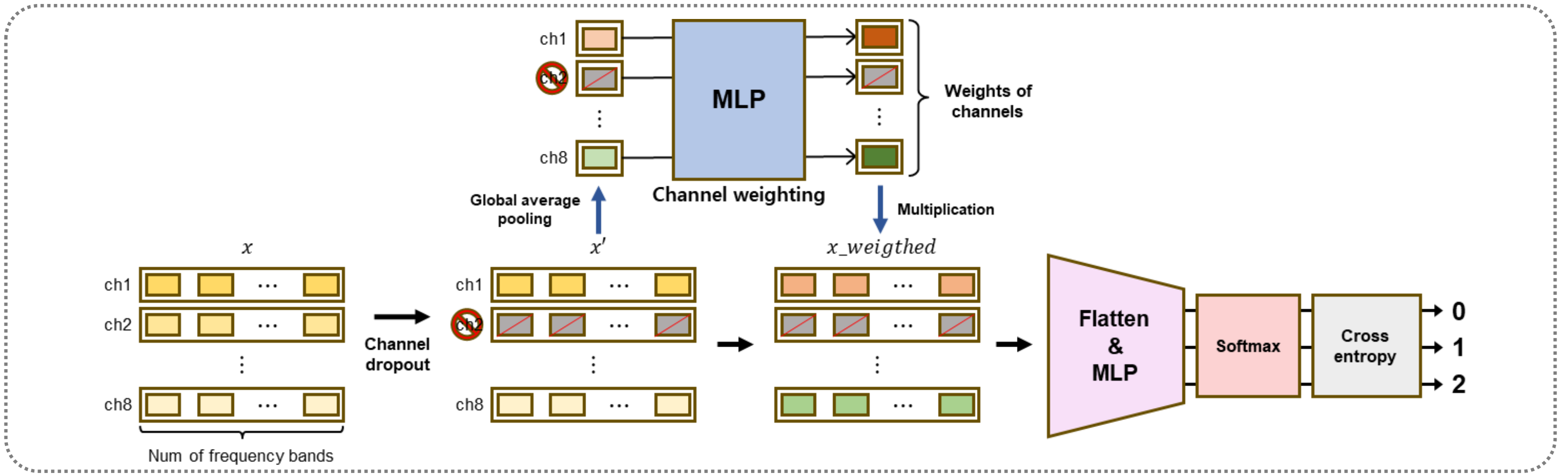
dongguk UNIVERSITY LINC 3.0
수인재두뇌과학 Suinjae Brain Science

- ADHD detection using EEG signals with a deep learning model

<CNN-based ADHD detection model>



<MLP-based ADHD detection model>



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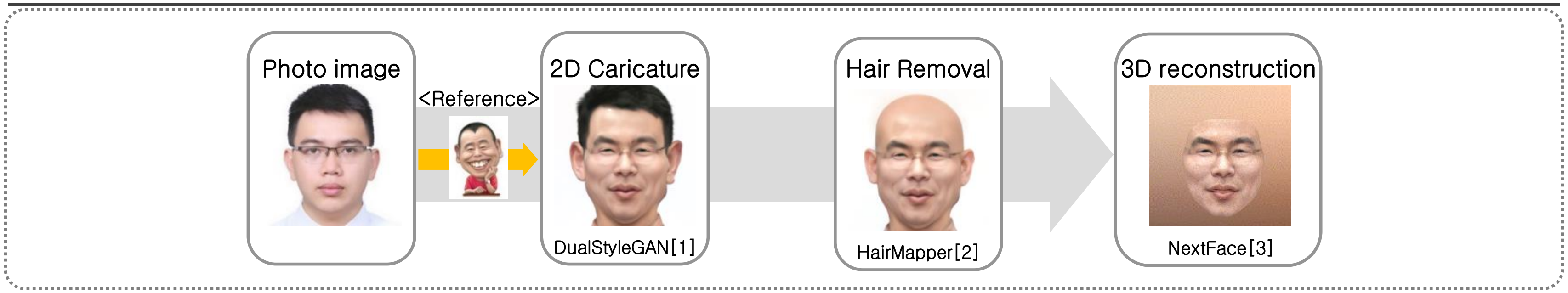
Applications of Artificial Intelligence

2022.04.01 ~ 2023.03.31 (Completed)

- Development of an automatic 3D caricature avatar creation and editing technology that emphasizes facial individuality

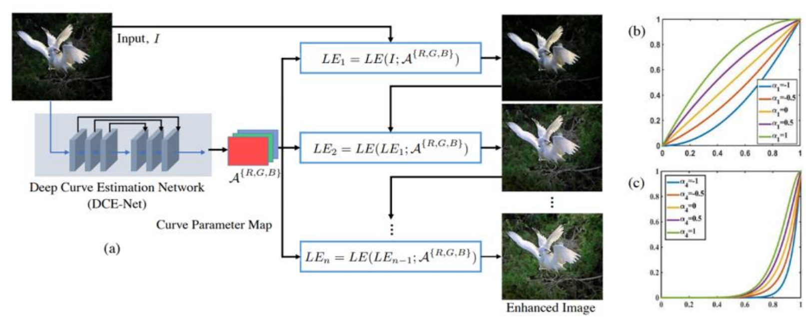
<Overall architecture>



- Development of techniques to enhance AR camera video recognition accuracy in nighttime lighting environments

<Experimental result>

<Low-light image enhancement model>



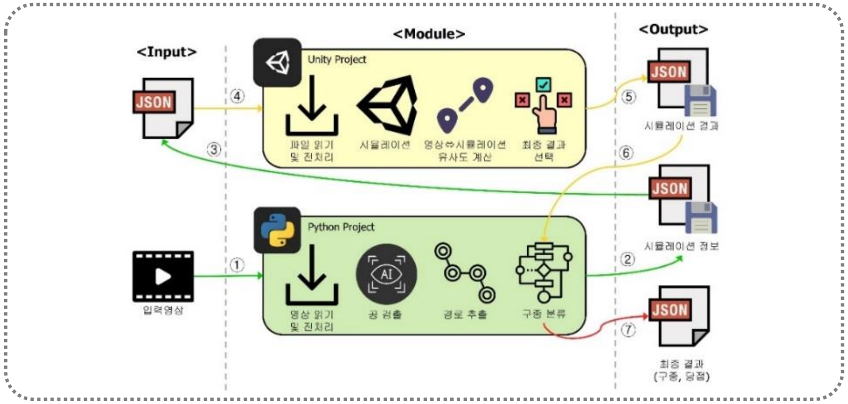
Applications of Artificial Intelligence

2022.03.10 ~ 2022.10.31 (Completed)
(주)한큐

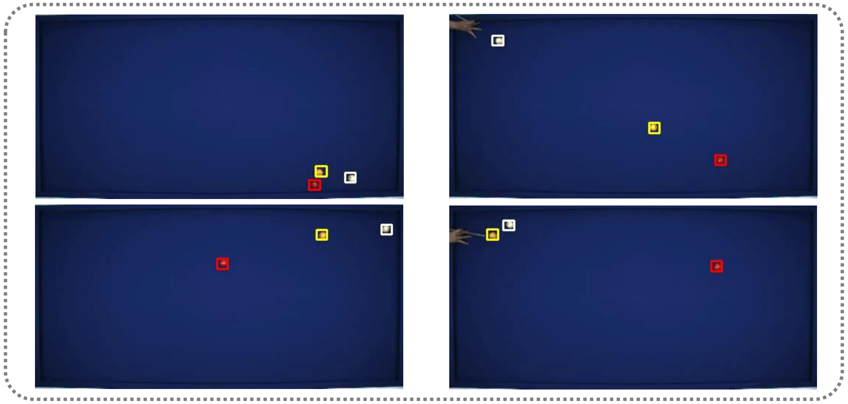
- 3-cushion billiards analysis

1. Situation analysis from input video (collisions, trajectories, scoring outcomes, etc.)
2. Deep learning/simulation–based prediction of cue-ball impact point and shot type

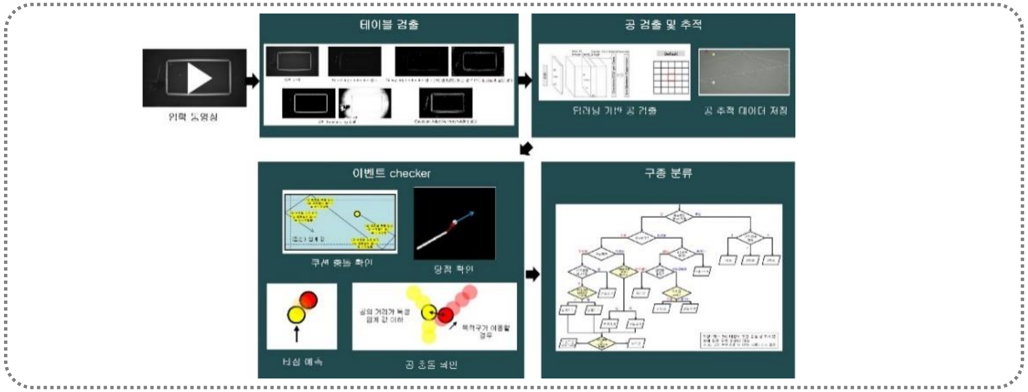
<Overall architecture>



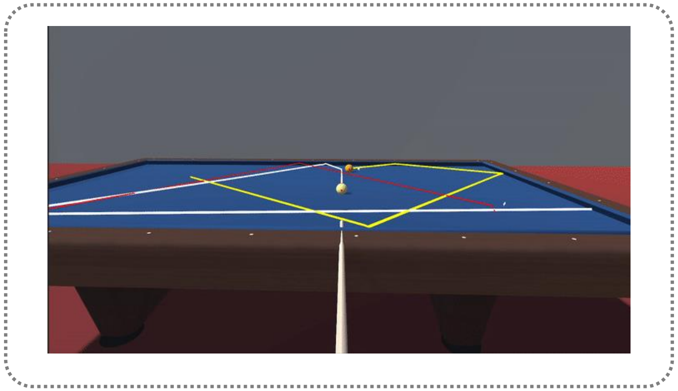
<Vision Recognition Example>



<Ball detection and shot-type analysis>



<Unity Simulation>



Applications of Artificial Intelligence

Smart Vision & Media Lab.

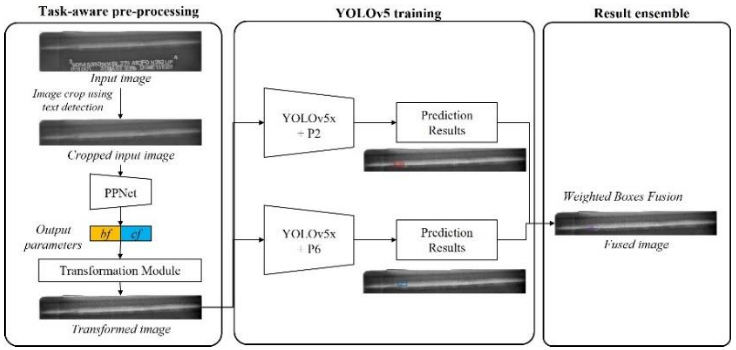
2022.04.01 ~ 2022.10.30 (Completed)

DSME 대우조선해양

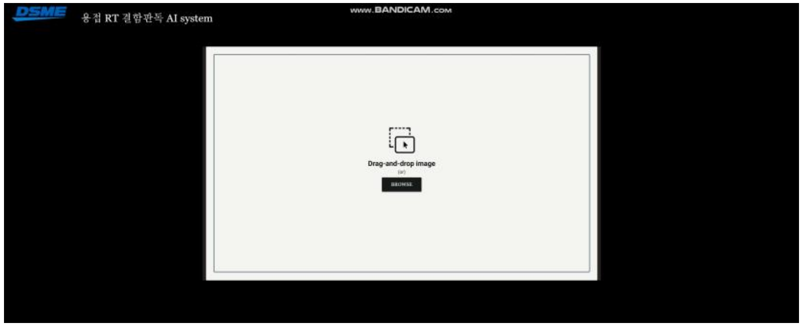
▪ Welding Defect Detection

1. Defect detection method applying adaptive image transformations to enhance defect image quality*

<Overall framework>

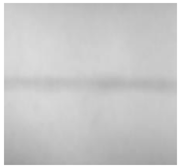


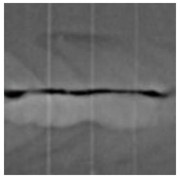
<Final video output of the defect detection system>

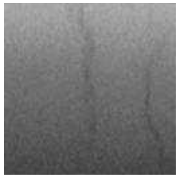


<Examples of weld defect types>

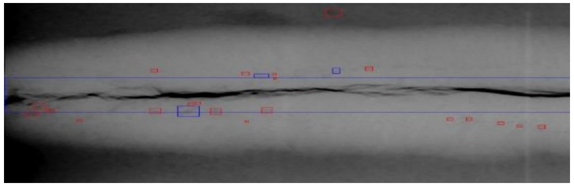

Porosity

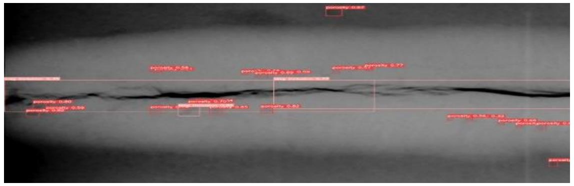

Slag inclusion


Undercut


Crack

<Example detection results>


Ground truth


Proposed

*J. E. Kwon, J. H. Park, J. H. Kim, Y. H. Lee, and S. I. Cho, "Context and Scale-aware YOLO for Welding Defect Detection," in NDT and E International, vol. 139, pp. 1-9, Oct. 2023.

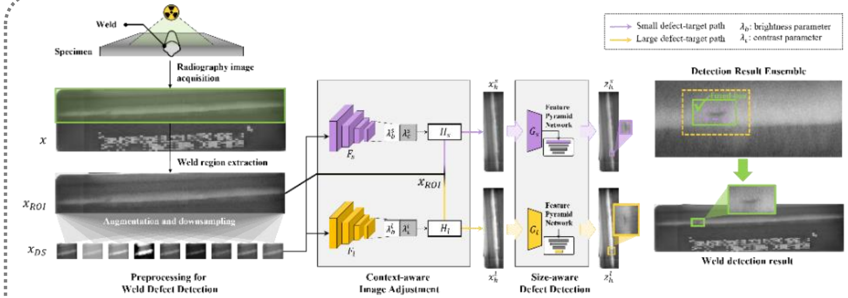
SOGANG UNIVERSITY

2022.04.01 ~ 2022.10.30 (Completed)

▪ Welding Defect Detection

1. Defect detection method applying adaptive image transformations to enhance defect image quality*

<Proposed pseudo-labeling of OD>

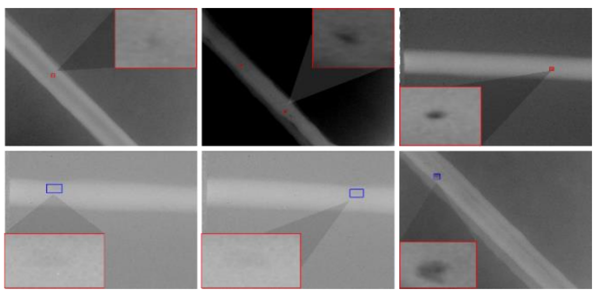


Key contributions:

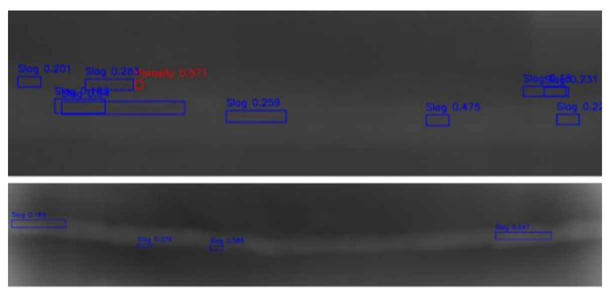
<Defect detection performance evaluation>

Method\Class	Precision (%)		Recall (%)		F1 score (%)	
	PO	SI	PO	SI	PO	SI
Faster R-CNN	52.2	42.9	60.4	51.4	56.0	46.8
Dynamic R-CNN	52.4	43.0	60.4	52.0	56.1	47.1
Cascade R-CNN	54.6	51.0	57.6	52.0	56.1	51.5
RetinaNet	50.0	44.2	51.1	40.1	50.5	42.1
YOLOv3	70.9	54.6	65.7	81.2	68.2	65.3
YOLOv5	62.8	58.3	74.6	**81.4**	68.2	67.9
DRepDet	50.6	50.4	44.0	41.5	47.1	45.5
MIG	71.9	47.4	65.5	83.6	68.6	60.5
LF-YOLO	66.7	73.1	67.2	63.9	66.9	68.2
Steel-pipe	64	55.1	74.4	80.1	68.8	65.3
Proposed method	**74.9**	**76.9**	**75.2**	78.3	**75.0**	**77.6**

<Examples of weld joint defects. Slag, Porosity>



<Sample detection results>



Defect detection results of the proposed method on the DSME dataset

Slag,
Porosity

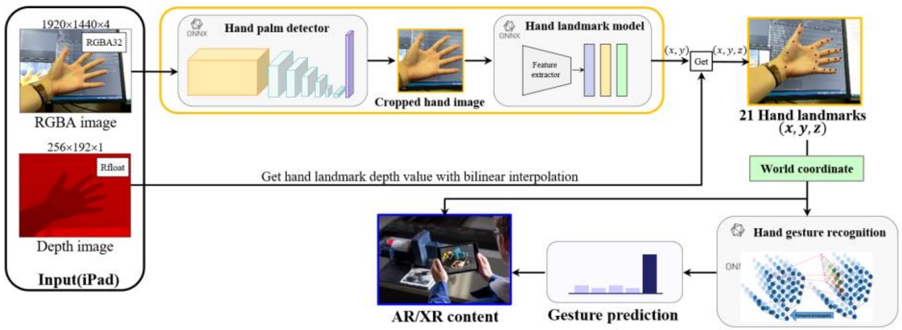
*J. E. Kwon, J. H. Park, J. H. Kim, Y. H. Lee, and S. I. Cho, "Context and Scale-aware YOLO for Welding Defect Detection," in NDT and E International, vol. 139, pp. 1-9, Oct. 2023.

Applications of Artificial Intelligence

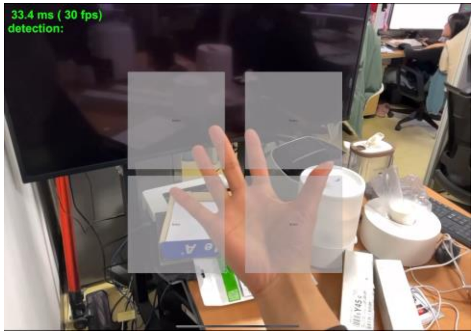
2022.01.01 ~ 2022.12.31 (Completed)

- Dynamic hand gesture recognition for user-interactive content (AR/XR)
  1. Development of a system for interacting with Unity AR/XR content using an iPad
  2. Hand landmark detection and hand gesture detection using hand landmarks

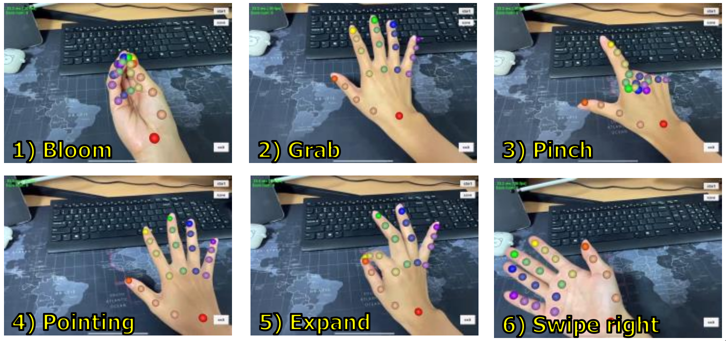
<Overall framework>



<Demo result>



<Hand gesture dataset acquisition>



Result of hand gesture detection

- Training
  : 540 samples × 6 = 3,240 samples

Index	Gesture	Accuracy
1	Bloom	1.0
2	Grab	1.0
3	Pinch	1.0
4	Pointing	0.9981
5	Expand	1.0
6	Swipe Right	0.9981

- Validation
  : 180 samples × 6 = 1,080 samples

Index	Gesture	Accuracy
1	Bloom	1.0
2	Grab	0.9948
3	Pinch	1.0
4	Pointing	0.9946
5	Expand	0.9885
6	Swipe Right	1.0

- Testing
  : 40 samples × 6 = 160 samples

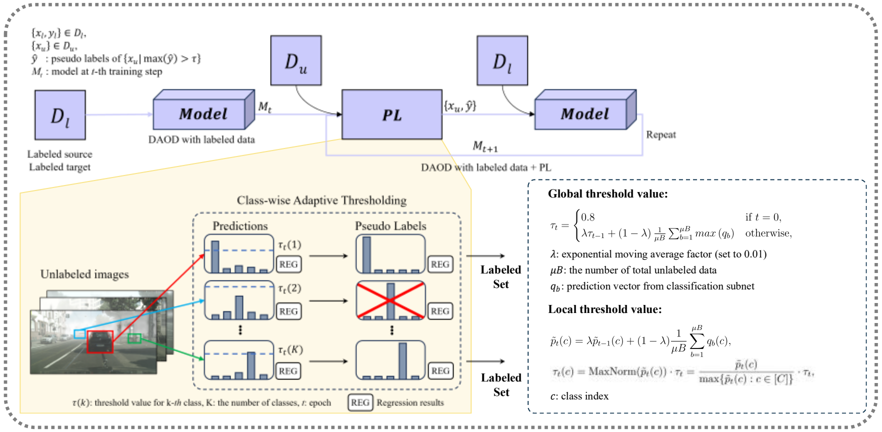
Index	Gesture	Accuracy
1	Bloom	1.0
2	Grab	0.975
3	Pinch	1.0
4	Pointing	0.975
5	Expand	1.0
6	Swipe Right	1.0

2023.03.01 ~ 2024.02.29 (Completed)

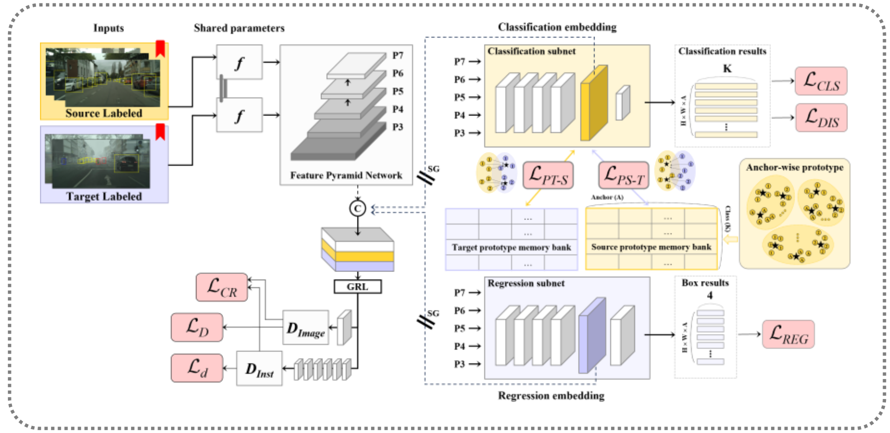
LG디스플레이

- Research on semi-supervised domain adaptation (SSDA) and object detection (OD) techniques for AI-based visual inspection
  1. Pseudo-labeling-based SSDA for OD
  2. Active learning for domain adaptive object detection (DAOD)

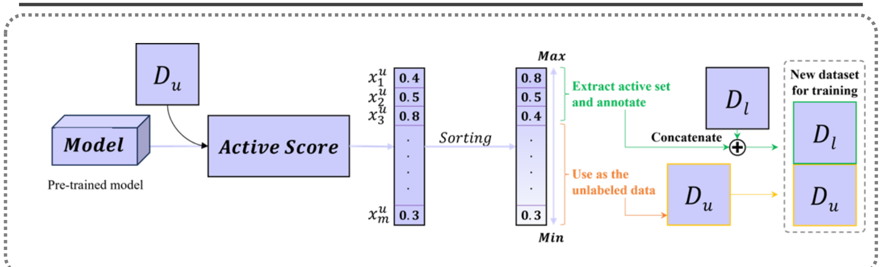
<Overall framework of pseudo-labeling-based SSDA for OD>



<Overall framework of baseline>



<Overall framework of proposed active learning for DAOD>



<DA performance when applying active learning>

	Experiment			Class-wise F1										F1
		#ALs	Portion	1	2	3	4	5	6	7	8	9	10	
	Without DA													
0	Fully-supervised			0.98	0.98	0.97	0.83	0.74	0.99	0.57	0.78	0.76	0.87	93.4
	With DA													
1	Baseline			0.96	0.81	0.97	0.70	0.70	0.87	0.67	0.82	0.68	0.42	88.6
	With PL													
2	FreeMatch			0.98	0.86	0.97	0.73	0.76	0.93	0.58	0.83	0.78	0.70	90.8
	With active learning													
3	(Baseline + AL 1) + FreeMatch	100	2.90%	0.98	0.98	0.97	0.83	0.74	0.99	0.57	0.78	0.76	0.87	92.2
	(Baseline + AL 2) + FreeMatch	500	14.49%	0.97	0.98	0.97	0.83	0.76	0.98	0.67	0.80	0.68	0.83	92.3
	(Baseline + AL 3) + FreeMatch	975	28.25%	0.98	0.98	0.98	0.82	0.62	0.98	0.42	0.78	0.74	0.70	92.7
	(Baseline + AL 4) + FreeMatch	2045	59.26%	0.98	0.98	0.97	0.83	0.77	0.99	0.55	0.77	0.72	0.85	**92.8**

B. H. Ngo, Y. J. Chae, S. J. Park, and S. I. Cho, "Multiple Tasks-based Multi-source Domain Adaptation Using Divide-and-conquer Strategy," in IEEE Access, Vol. 11, pp. 134969 –134985 , Dec. 2023.
B. H. Ngo, J. H. Kim, S. J. Park and S. I. Cho, "Collaboration Between Multiple Experts for Knowledge Adaptation on Multiple Remote Sensing Sources," in IEEE Transactions on Geoscience and Remote Sensing, vol. 60, pp. 1-15, Jul. 2022.
J. H. Kim, B. H. Ngo, J. H. Park, J. E. Kwon, H. S. Lee, and S. I. Cho, "Distilling and Refining Domain-Specific Knowledge for Semi-Supervised Domain Adaptation," BMVC, London, UK, Nov. 21~24, 2022.
B. H. Ngo, J. H. Kim, Y. J. Chae and S. I. Cho, "Multi-View Collaborative Learning for Semi-Supervised Domain Adaptation," in IEEE Access, Vol. 9, pp. 166488-166501, Dec. 2021.
B. H. Ngo, J. H. Park, S. J. Park, and S. I. Cho, "Semi-supervised Domain Adaptation using Explicit Class-wise Matching for Domain-Invariant and Class-Discriminative Feature Learning," in IEEE Access, Vol. 9, pp. 128467-128480, Sep. 2021.

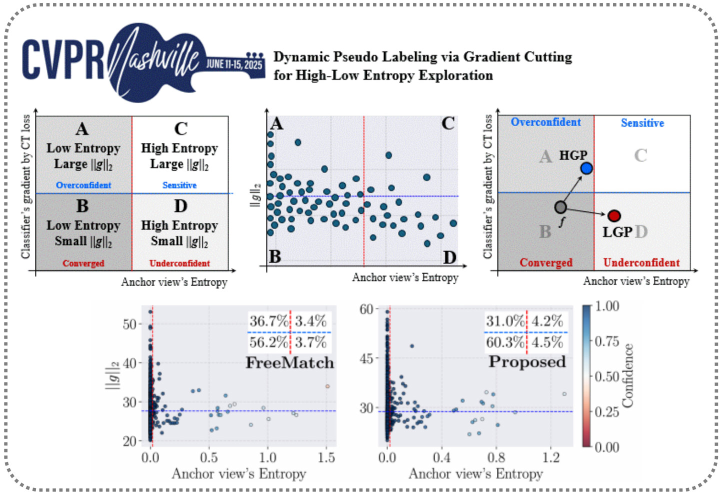
Smart Vision & Media Lab.

2025.06.11~2025.06.14 (Published)

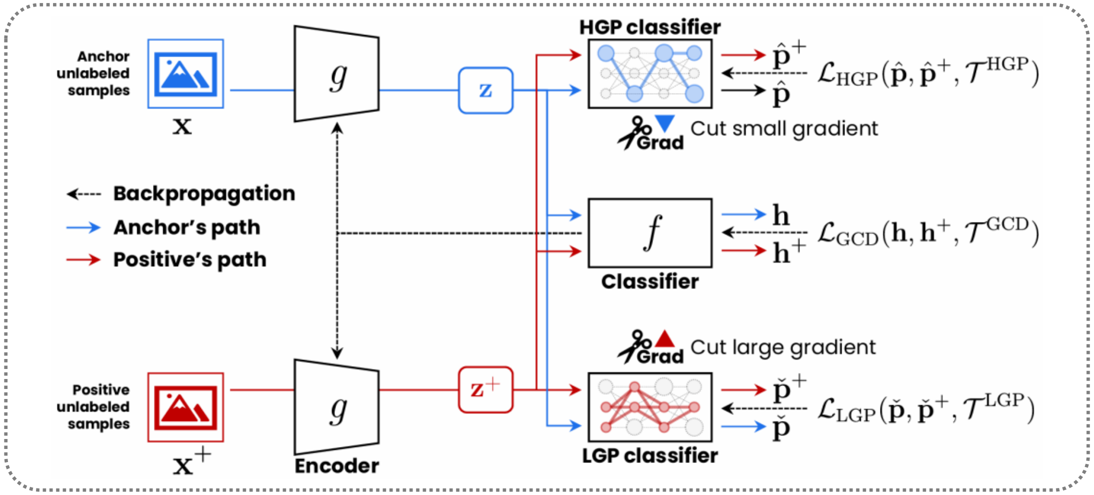
- Semi-supervised Learning* (CVPR2025, BK IF=4, h5-index=440)

1.Dynamic pseudo labeling via gradient cutting for high-low entropy exploration

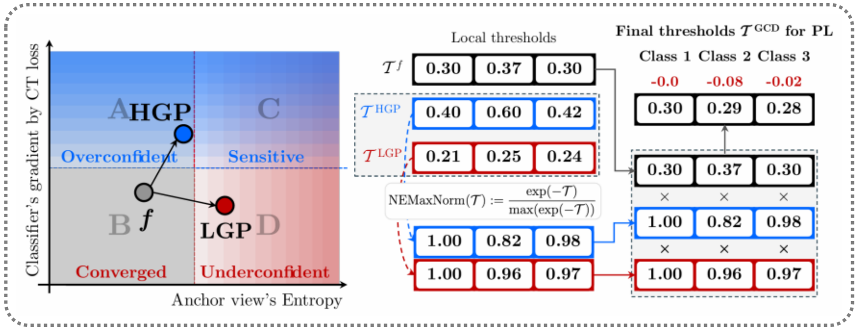
<Visualization of the four-learning status>



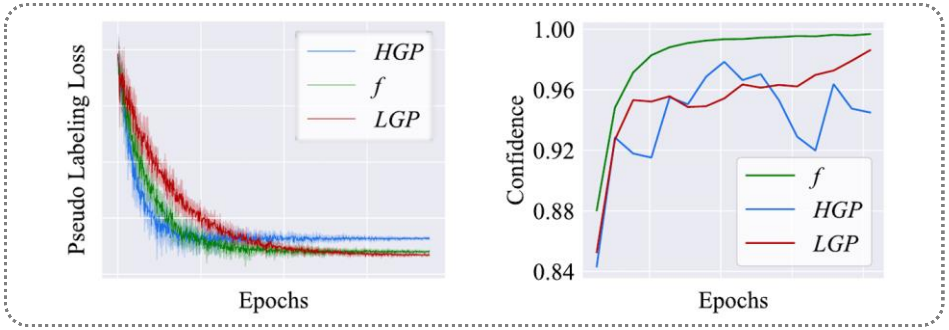
<Overall architecture of proposed method>



<Flowchart of the adjustment of threshold>



<Training trajectories of the proposed method>



*Park, Jae Hyeon, et al. "Dynamic Pseudo Labeling via Gradient Cutting for High-Low Entropy Exploration." Proceedings of the Computer Vision and Pattern Recognition Conference. 2025.

2025.06.11~2025.06.14 (Published)

- Semi-supervised Learning* (CVPR2025, BK IF=4, h5-index=440)

1. Dynamic pseudo labeling via gradient cutting for high-low entropy exploration

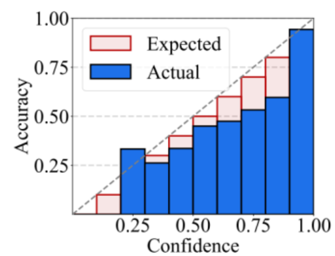
<Main result>

#	Dataset / Method	Venue	CIFAR-10 40	CIFAR-10 250	CIFAR-10 4,000	CIFAR-100 400	CIFAR-100 2,500	CIFAR-100 10,000	SVHN 40	SVHN 250	SVHN 1,000	STL-10 40	STL-10 250	STL-10 1,000
1	Fully-supervised			4.57±0.06			18.96±0.06			2.14±0.01				
2	Pseudo-Labeling [18]	ICMLW'13	76.29±1.08	48.28±2.10	14.90±0.20	87.15±0.87	59.09±0.61	38.86±0.09	75.95±3.39	16.60±1.13	9.33±0.58			
3	FixMatch [32]	NIPS'20	11.39±3.35	5.07±0.33	4.31±0.15	48.85±1.75	28.29±0.11	22.60±0.12	3.96±2.17	2.48±0.38	2.28±0.11			5.17±0.63
4	UDA [38]	NIPS'20	8.01±1.34	5.43±0.96	4.32±0.08	53.44±2.06	34.27±0.28	27.52±0.10		2.72±0.40	2.23±0.07			
5	MPL [29]	CVPR'21			**3.89±0.07**						1.99±0.07			
6	DASH [39]	ICML'21	13.22±3.75	**4.56±0.13**	4.08±0.06	44.76±0.96	27.18±0.21	21.97±0.14	3.03±1.59	2.17±0.10	2.03±0.06			7.26±0.40
7	FlexMatch [43]	NIPS'21	4.97±0.06	4.98±0.09	4.19±0.01	39.94±1.62	26.49±0.20	21.90±0.15	8.19±3.20	5.02±1.20	6.72±0.30	29.15±4.16	8.23±0.39	5.77±0.18
8	CoMatch [19]	CVPR'21	6.91±1.39	4.91±0.33	4.29±0.04				9.51±5.59	2.21±0.20	1.96±0.07			20.20±0.38
9	AdaMatch [31]	ICLR'22	5.05±0.04	5.01±0.11	4.29±0.06	47.82±1.48	33.26±0.55	27.53±0.12	2.32±0.17	2.14±0.10	2.04±0.02			
10	SimMatch [46]	CVPR'22	5.60±1.37	4.84±0.39	3.96±0.01	37.81±2.21	25.17±0.32	20.58±0.11	3.19±0.74	2.26±0.11	2.08±0.06			
11	ConMatch [15]	ECCV'22	**4.43±0.13**	4.70±0.25	3.92±0.08	38.89±2.18	25.39±0.20		3.14±0.57	3.13±0.72				5.26±0.04
12	SoftMatch [3]	ICLR'23	4.91±0.12	4.82±0.09	4.04±0.02	37.10±0.77	26.66±0.25	22.03±0.03	2.33±0.25		2.01±0.01	21.42±3.48		5.73±0.24
13	ReFixMatch [26]	ICCV'23	4.94±0.01	4.83±0.05	4.18±0.05	46.12±1.07	27.28±0.22	21.60±0.04	2.15±1.23		1.89±0.03	28.60±4.12	8.21±0.30	5.74±0.30
14	CRMatch [9]	IJCV'23	5.52±0.32	5.21±0.06	4.26±0.19	**35.72±0.50**	**24.61±0.37**	**20.91±0.24**	2.79±0.93	2.35±0.29	2.08±0.07			**4.89±0.17**
15	FreeMatch [37]	ICLR'23	4.90±0.04	4.88±0.18	4.10±0.02	37.98±0.42	26.47±0.20	21.68±0.03	**1.97±0.02**	1.97±0.01	1.96±0.03	15.56±0.55		5.63±0.15
16	SimMatchV2 [47]	ICCV'23	4.90±0.16	5.04±0.09	4.33±0.16	36.68±0.86	26.66±0.38	21.37±0.20	7.92±2.80	2.92±0.81	2.85±0.91	15.85±2.62	7.54±0.81	5.65±0.26
17	RelationMatch [45]	ICLR'24	4.96±0.05	4.88±0.05	4.17±0.04	39.89±1.43	26.48±0.18	21.88±0.16				13.94±3.76	8.16±0.34	5.68±0.19
18	EPASS w/ [46]	WACV'24	5.55±0.21	5.31±0.13	4.23±0.05	50.73±0.33	29.51±0.16	22.16±0.12	2.98±0.02	**1.93±0.05**	**1.85±0.04**	**9.15±3.25**	**6.27±0.03**	5.40±0.12
	EPASS w/ [15]		5.31±0.10	5.08±0.05	4.37±0.03	38.88±0.24	25.68±0.33	21.32±0.14	2.31±0.04	2.04±0.02	2.02±0.02	15.71±2.48	8.08±0.26	5.58±0.04
	Ours	-	**4.57±0.05**	**4.46±0.07**	**3.91±0.01**	**31.84±0.75**	**24.95±0.15**	**20.20±0.07**	**2.01±0.03**	**1.90±0.01**	**1.81±0.00**	**13.5±2.01**	**7.47±0.52**	**5.11±0.09**

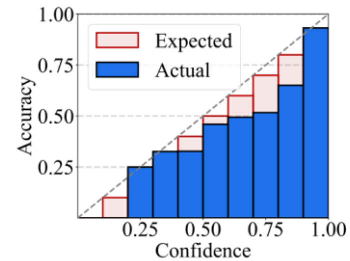
<Ablation study for proposed components>

$\mathcal{T}^f$	$\mathcal{T}^{HGP}$	$\mathcal{T}^{LGP}$	$\mathcal{T}^{GCD}$	CIFAR-10 (40)	CIFAR-100 (400)	SVHN (40)	STL-10 (40)
✓				10.04	38.42	**2.23**	**13.71**
✓	✓		✓	8.91	37.82	2.37	16.23
✓		✓	✓	**7.45**	**33.51**	2.41	15.43
✓	✓	✓	✓	**4.52**	**31.09**	**1.98**	**11.4**

<Reliability diagram>



FreeMatch (ECE 0.034)



Ours (ECE 0.027)

*Park, Jae Hyeon, et al. "Dynamic Pseudo Labeling via Gradient Cutting for High-Low Entropy Exploration." Proceedings of the Computer Vision and Pattern Recognition Conference. 2025.

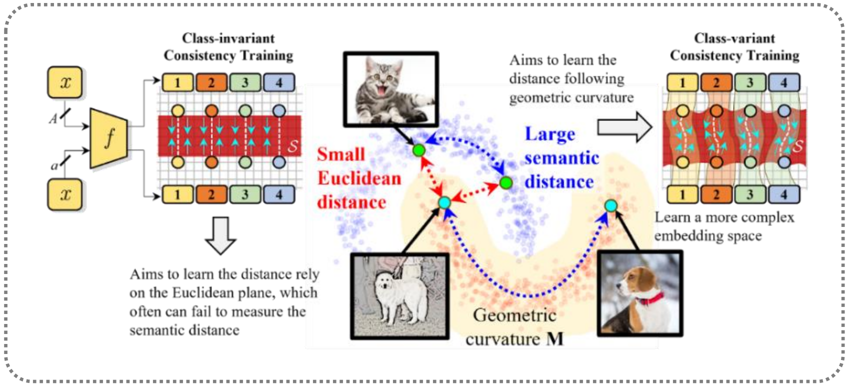
Applications of Artificial Intelligence

Smart Vision & Media Lab.

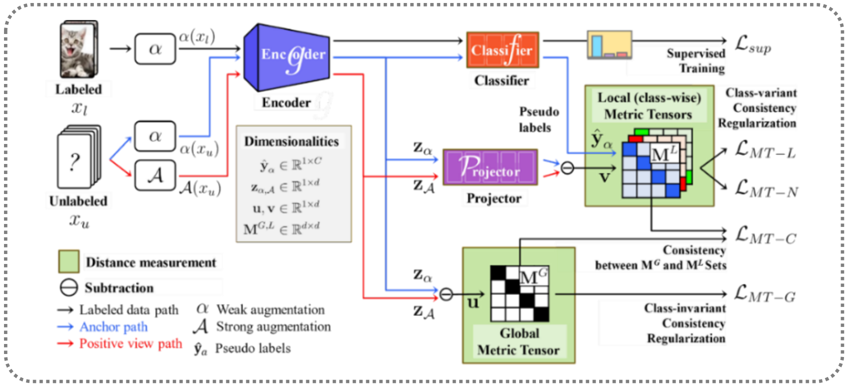
2024.06.17~2024.06.21 (Published)

- Semi-supervised Learning* (CVPR2024, BK IF=4, h5-index=440)
  1. Metric tensor-based semi-supervised learning (SSL) technique for representing geometric information in the embedding space

<Conceptual diagram of the motivation>



<Overall framework>



< Experimental result on CIFAR 10 and CIFAR 100 >

Dataset / Method	CIFAR-10 40	CIFAR-10 250	CIFAR-10 4,000	CIFAR-10 Avg.	CIFAR-100 400	CIFAR-100 2,500	CIFAR-100 10,000	CIFAR-100 Avg.
Fully-supervised		*4.91±0.33		-		*19.27±0.05		-
Pseudo-Labeling [19]	74.61±0.26	46.49±2.20	15.08±0.19	45.39±0.88	87.45±0.85	57.74±0.28	36.55±0.24	60.58±0.45
Mean Teacher [31]	70.09±1.60	37.46±3.30	8.10±0.21	38.55±1.70	81.11±1.44	45.17±1.06	31.75±0.23	52.67±0.91
VAT [21]	74.66±2.12	41.03±1.79	10.51±0.12	42.07±1.34	85.20±1.40	46.84±0.79	32.14±0.19	54.73±0.79
UDA [36]	10.62±3.75	5.15±0.06	4.29±0.07	6.69±1.29	46.39±1.59	27.73±0.21	22.49±0.23	32.20±0.67
DASH [37]	9.16±4.31	**4.56±0.13**	4.08±0.06	5.93±1.50	44.76±0.96	27.18±0.21	21.97±0.14	31.30±0.43
FixMatch [30]	7.47±0.28	*5.07±0.65	*4.26±0.05	5.60±0.33	*48.85±1.75	*28.29±0.11	*22.60±0.12	33.25±0.66
ReMixMatch [3]	9.88±1.03	6.30±0.05	4.84±0.01	7.01±0.36	42.75±1.05	26.03±0.35	**20.02±0.27**	29.60±0.56
FlexMatch [41]	4.97±0.06	4.98±0.09	4.19±0.01	4.71±0.05	39.94±1.62	26.49±0.20	21.90±0.15	29.44±0.65
SimMatch [43]	5.60±1.37	4.84±0.39	3.96±0.01	4.80±0.59	37.81±2.21	**25.07±0.32**	20.58±0.11	**27.82±0.87**
ConMatch [16]	**4.43±0.13**	**4.70±0.25**	**3.92±0.08**	**4.34±0.15**	38.89±2.18	25.39±0.20	-	-
SoftMatch [6]	4.91±0.12	4.82±0.09	4.04±0.02	4.59±0.07	37.10±0.77	26.66±0.25	22.03±0.03	28.60±0.35
ShrinkMatch [39]	5.08±0.18	4.74±0.25	-	-	**35.36±1.04**	**25.17±0.20**	-	-
Proposed	**4.37±0.21**	4.71±0.06	**3.92±0.09**	**4.33±0.12**	**37.49±1.92**	25.43±0.07	**20.18±0.25**	**27.70±0.74**

< Experimental result on SVHN and STL-10 >

Dataset / Method	SVHN 40	SVHN 250	SVHN 1,000	SVHN Avg.	STL-10 40	STL-10 250	STL-10 1,000	STL-10 Avg.
Fully-supervised		*2.13±0.01		-		None		-
Pseudo-Labeling [19]	64.61±5.60	15.59±0.95	9.40±0.32	29.87±2.29	74.68±0.99	55.45±2.43	32.64±0.71	54.26±1.37
Mean Teacher [31]	36.09±3.98	3.45±0.03	3.27±0.05	14.27±1.35	71.72±1.45	56.49±2.75	33.90±1.37	54.04±1.85
VAT [21]	74.75±3.38	4.33±0.12	4.11±0.20	27.73±1.23	74.74±0.38	56.42±1.97	37.95±1.12	56.37±1.15
UDA [36]	5.12±4.27	**1.92±0.05**	**1.89±0.01**	2.98±1.44	37.42±8.44	9.72±1.15	6.64±0.17	17.93±3.25
DASH [37]	3.03±1.59	2.17±0.10	2.03±0.06	2.41±0.58	-	-	**3.96±0.25**	-
FixMatch [30]	*3.96±2.17	*2.48±0.38	*2.28±0.11	2.91±0.89	35.97±4.14	9.81±1.04	6.25±0.33	17.34±1.83
ReMixMatch [3]	24.04±9.13	6.36±0.22	5.16±0.31	11.85±3.22	32.12±6.24	12.49±1.28	6.74±0.14	17.12±2.55
FlexMatch [41]	8.19±3.20	6.59±2.29	6.72±0.30	7.17±1.93	29.15±4.16	**8.23±0.39**	5.77±0.18	14.38±1.57
ConMatch [16]	3.14±0.57	3.13±0.72	-	-	-	-	**5.26±0.04**	-
SoftMatch [6]	**2.33±0.25**	*2.21±0.00	**2.01±0.01**	**2.18±0.08**	21.42±3.48	*9.22±0.01	5.73±0.24	**12.12±1.24**
ShrinkMatch [39]	2.51±0.56	**1.96±0.04**	-	-	**14.02±0.41**	8.45±0.62	-	-
Proposed	**2.13±0.26**	2.14±0.06	2.04±0.04	**2.10±0.12**	**20.47±5.44**	**8.45±0.30**	5.27±0.11	**11.39±1.95**

*J. H. Park, G. Lee, S. Park, and S. I. Cho, "Not All Classes Stand on Same Embeddings: Calibrating a Semantic Distance with Metric Tensor," In Proceedings of the IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR2024), 2024.

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Applications of Artificial Intelligence

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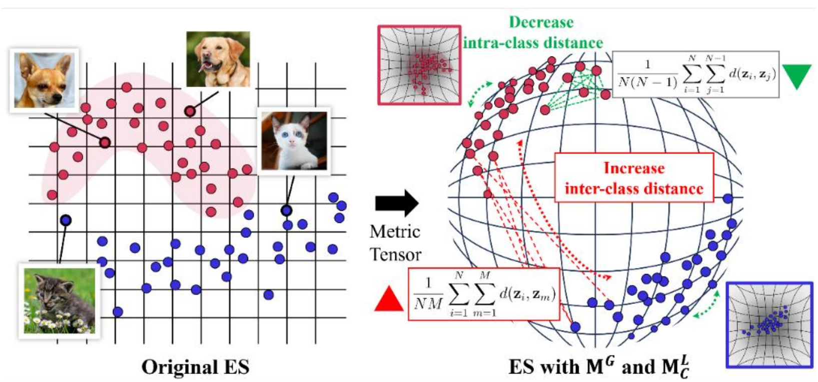
2024.06.17~2024.06.21 (Published)

- Semi-supervised Learning* (CVPR2024, BK IF=4, h5-index=440)

1. Metric tensor-based semi-supervised learning (SSL) technique for representing geometric information in the embedding space

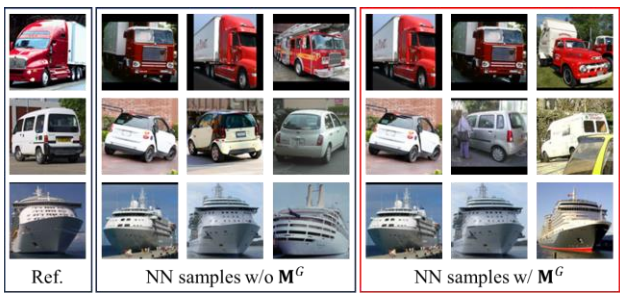
<Conceptual diagram of the proposed metric-tensor-based consistency regularization>



<Supervised learning experiment results>

Architecture	Accuracy (%)	Param.	Resolution
ViT-B	75.26	88M	224^2
ViT-L	75.56	307M	224^2
ViT-B + MT	76.84 (+1.58)	-	224^2
ViT-L + MT	78.24 (+2.68)	-	224^2
Swin-S	82.81	50M	224^2
Swin-B	83.18	88M	224^2
Swin-S + MT	84.10 (+1.29)	-	224^2
Swin-B + MT	84.50 (+1.32)	-	224^2
Swin-T-v2	81.50	28M	256^2
Swin-S-v2	83.18	50M	256^2
Swin-B-v2	83.69	88M	256^2
Swin-T-v2 + MT	82.87 (+1.37)	-	256^2
Swin-S-v2 + MT	84.79 (+1.61)	-	256^2
Swin-B-v2 + MT	85.23 (+1.54)	-	256^2

<Nearest neighbor samples extraction results>



<Distance and similarity using the metric tensor>

두 샘플의 의미론적 거리 도출

$$s^2 = (p_1 - q_1)^2 + (p_2 - q_2)^2,$$
$$= (\mathbf{p} - \mathbf{q})(\mathbf{p} - \mathbf{q})^\top,$$
$$= \mathbf{u}\mathbf{u}^\top,$$
$$s^2 = \mathbf{u}\mathbf{M}\mathbf{u}^\top,$$

두 샘플의 의미론적 유사도 attention 도출

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V,$$
$$\downarrow$$
$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{D(Q, K)}{\sqrt{d_k}})V,$$
$$D = \sqrt{(Q-K)\mathbf{M}(Q-K)},$$

*J. H. Park, G. Lee, S. Park, and S. I. Cho, "Not All Classes Stand on Same Embeddings: Calibrating a Semantic Distance with Metric Tensor," In Proceedings of the IEEE / Computer Vision and Pattern Recognition Conference (CVPR2024), 2024.

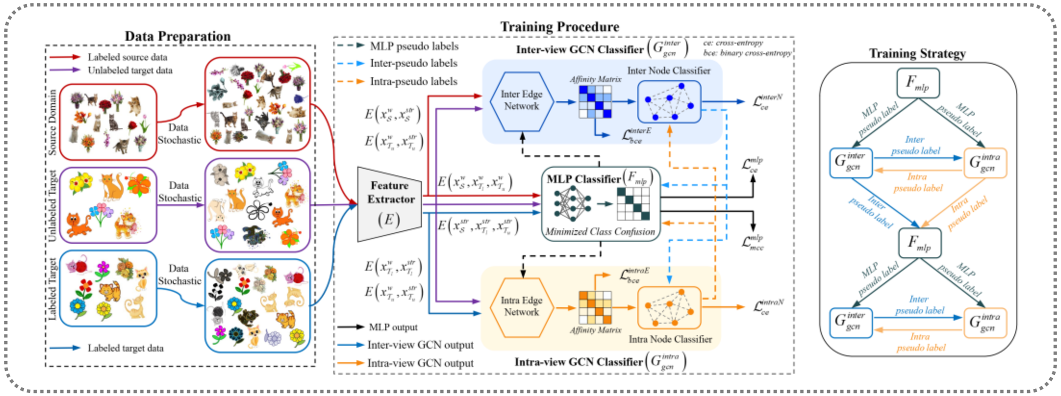
2023.10.04~2023.10.06 (Published)

ICCV23 PARIS

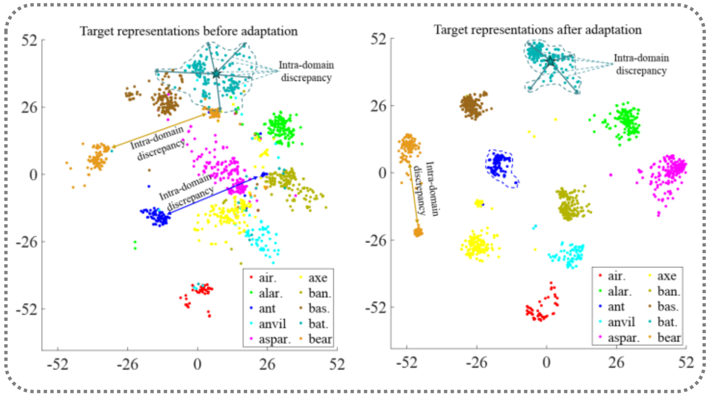
- SSDA* (ICCV2023, BK IF=4, Top-tier Conference, h5-index=291)

1. Domain adaptation method using the collaboration of three training techniques
   - Application of inter- and intra-view-based pseudo-labeling among three types of classifiers

<Overall framework>



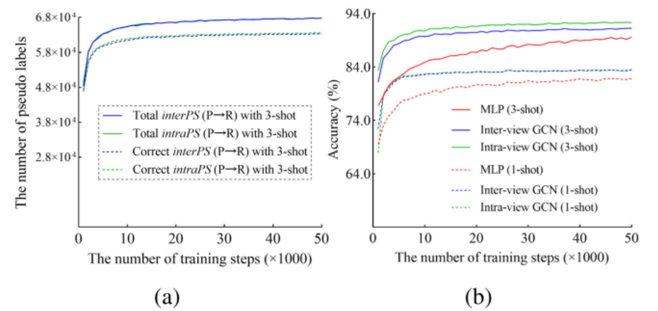
<Conceptual diagram of inter- and intra-domain>



<Main result>

Net	Method	R→C	R→P	R→A	P→R	P→C	P→A	A→P	A→C	A→R	C→R	C→A	C→P	Mean
AlexNet	PAC	58.9	72.4	47.5	61.9	53.2	39.6	63.8	49.9	60.0	54.5	36.3	64.8	55.2
	APE	51.9	74.6	51.2	61.6	47.9	42.1	65.5	44.5	60.9	58.1	44.3	64.8	55.6
	Con^{2}DA	52.3	73.5	49.1	64.4	49.3	38.2	66.4	47.7	62.4	59.9	39.9	66.1	55.8
	CDAC	54.9	75.8	51.8	64.3	51.3	43.6	65.1	47.5	63.1	63.0	44.9	65.6	56.8
	MVCL	55.4	73.1	54.6	65.6	49.9	44.7	66.0	47.9	64.5	59.7	42.9	63.3	57.3
	CLDA	51.5	74.1	54.3	67.0	47.9	47.0	65.8	47.4	66.6	64.1	46.8	67.5	58.3
	ECACL	55.4	75.7	56.0	67.0	52.5	46.4	67.4	48.5	66.3	60.8	45.9	67.3	59.1
	TriCT	**59.9**	**84.2**	**61.8**	**76.6**	**55.4**	**50.6**	**75.3**	**55.4**	**74.4**	**69.6**	**49.6**	**76.3**	**65.8**
ResNet-34	MME	64.6	85.5	71.3	80.1	64.6	65.5	79.0	63.6	79.7	76.6	67.2	79.3	73.1
	APE	66.4	86.2	73.4	82.0	65.2	66.1	81.1	63.9	80.2	76.8	66.6	79.9	74.0
	CDAC	67.8	85.6	72.2	81.9	67.0	67.5	80.3	65.9	80.6	80.2	67.4	81.4	74.2
	CLDA	66.0	87.6	76.7	82.2	63.9	72.4	81.4	63.4	81.3	80.3	70.5	80.9	75.5
	MVCL	69.6	88.1	76.4	80.9	66.0	71.2	82.0	66.0	79.6	79.6	67.4	80.7	75.6
	DECOTA	70.4	87.7	74.0	82.1	68.0	69.9	81.8	64.0	80.5	79.0	68.0	83.2	75.7
	MCL	70.1	88.1	75.3	83.0	68.0	69.9	83.9	67.5	82.4	81.6	71.4	84.3	77.1
	TriCT	**81.9**	**94.1**	**86.3**	**92.3**	**78.7**	**83.4**	**91.1**	**76.9**	**91.3**	**91.8**	**80.5**	**92.5**	**86.7**

<Quality evaluation of pseudo label>



*B. H. Ngo, Y. J. Chae, J. E. Kwon, J. H. Park, and S. I. Cho, "Improved Knowledge Transfer for Semi-supervised Domain Adaptation via Trico Training Strategy," In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV2023) 2023.

Applications of Artificial Intelligence

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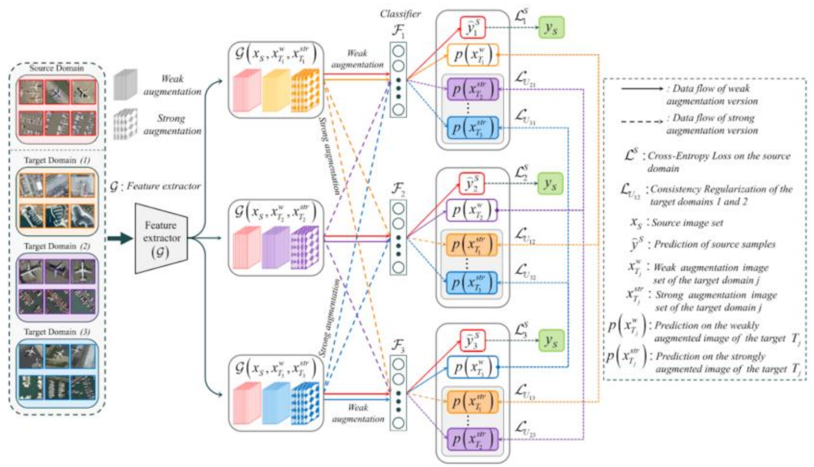
2023 (Published)

IEEE

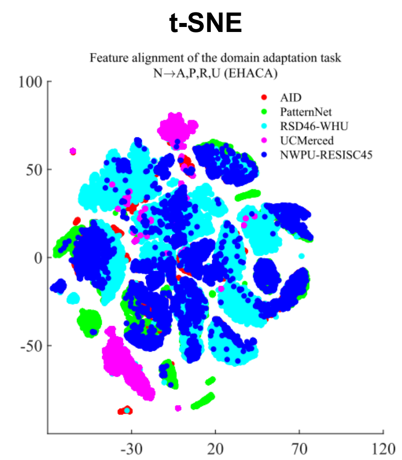
- Multitarget Domain Adaptation* (ITGRS 2023 [JCR 4.5%, IF: 7.5])

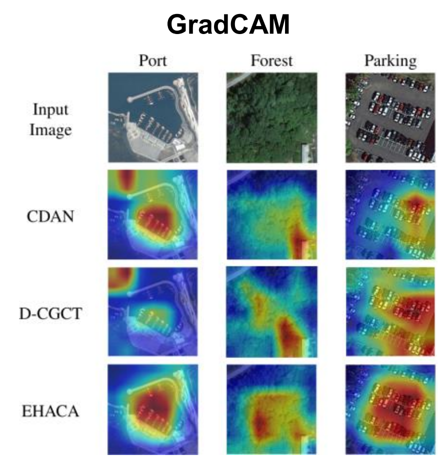
1. Domain adaptation technique for transfer learning from a single source domain to multiple target domains

<Overall framework>



<Qualitative result>





<DA performance (Backbone: AlexNet)>

Setting	Source / Target	A→N	A→P	A→R	A→U	N→A	N→P	N→R	N→U	P→A	P→N	P→R	P→U	R→A	R→N	R→P	R→U	U→A	U→N	U→P	U→R	Avg. (%)
S–T	DAN	80.8	80.5	67.2	72.3	86.8	87.9	58.4	82.3	66.2	68.5	50.5	82.7	69.0	60.9	67.6	62.0	67.0	61.8	91.2	50.0	70.7
	DANN	83.8	84.3	70.8	82.6	86.7	89.4	62.2	84.8	68.2	**72.8**	**55.0**	92.9	74.1	65.1	**72.6**	**69.3**	67.7	65.6	**92.4**	47.7	74.4
	JAN	82.5	**90.8**	70.5	75.2	86.0	**91.3**	56.2	83.6	71.6	70.1	54.7	89.3	70.9	64.9	64.7	67.4	**72.3**	69.2	91.3	**52.9**	73.8
	CDAN	**86.3**	88.4	**77.9**	**86.3**	**91.0**	89.2	**69.6**	**90.6**	**73.5**	71.6	50.2	**95.0**	**75.9**	**68.5**	71.9	68.8	68.6	**71.0**	91.2	50.4	**76.8**
T–C	DAN	73.1				69.3				54.3				62.9				58.2				63.6
	DANN	75.0				71.3				57.2				66.5				61.4				66.3
	JAN	74.8				69.4				**62.8**				64.2				**67.6**				67.8
	CDAN	**76.1**				**75.7**				59.7				**67.6**				63.5				**68.5**
	TSAN	70.6				69.2				53.9				64.4				58.2				63.3
M–T	D-CGCT	90.4	90.8	76.7	**96.6**	**93.1**	87.4	71.6	86.8	64.8	56.5	42.7	75.4	78.7	78.8	**92.0**	**81.3**	73.8	68.1	92.3	50.6	77.4
	EHACA	**95.4**	**91.4**	**93.0**	94.5	92.7	**99.8**	**92.0**	**96.1**	**87.1**	**84.2**	**88.5**	**97.9**	**83.0**	**79.8**	84.8	80.0	**85.3**	**84.3**	**99.8**	**83.3**	**89.6**

Note: S–T: Single-Target; T–C: Target-Combined; M–T: Multi-Target

<DA performance (Backbone: ResNet50)>

Setting	Source / Target	A→N	A→P	A→R	A→U	N→A	N→P	N→R	N→U	P→A	P→N	P→R	P→U	R→A	R→N	R→P	R→U	U→A	U→N	U→P	U→R	Avg. (%)
S–T	DAN	84.2	82.6	68.5	77.6	89.0	92.2	65.9	82.9	66.5	69.3	52.9	83.4	69.2	64.9	67.7	63.8	68.3	62.5	94.0	53.4	72.9
	DANN	89.4	90.1	77.9	92.1	92.7	91.0	69.2	88.4	**77.9**	75.6	55.7	94.5	80.9	78.9	80.5	72.9	81.0	73.7	**93.9**	53.9	80.5
	JAN	88.1	91.0	72.2	77.8	89.4	92.0	66.8	86.0	77.0	**79.8**	**59.3**	90.9	73.3	76.0	75.9	70.3	75.6	**78.5**	92.5	**62.3**	78.7
	CDAN	**95.0**	**97.4**	**81.2**	**92.3**	**97.1**	**93.3**	**84.1**	**96.2**	**77.9**	78.5	58.3	**98.1**	**83.4**	**79.7**	**81.2**	**73.3**	**85.5**	75.1	93.1	57.0	**83.9**
T–C	DAN	76.3				74.0				55.1				68.0				60.6				66.8
	DANN	79.7				76.5				61.7				74.8				63.3				71.2
	JAN	78.5				73.0				**75.3**				69.4				71.6				73.5
	CDAN	84.1				79.1				67.9				**76.6**				**71.7**				**75.9**
	TSAN	**84.3**				**81.9**				60.5				73.4				68.1				73.6
M–T	D-CGCT	96.6	99.3	84.2	**98.9**	**98.1**	97.5	81.0	**98.9**	80.2	76.3	69.5	91.8	91.2	84.1	92.4	85.9	88.5	77.7	96.6	75.4	88.2
	EHACA	**97.7**	**99.9**	**97.0**	97.6	97.2	**99.9**	**97.1**	98.8	**89.9**	**90.2**	**93.0**	**98.4**	**97.0**	**96.0**	**99.6**	**97.3**	**97.2**	**90.4**	**99.8**	**91.5**	**96.3**

*B. H. Ngo, Y. J. Chae, J. H. Park, and S. I. Cho, "Easy-to-Hard Structure for Remote Sensing Scene Classification in Multitarget Domain Adaptation," in IEEE Transactions on Geoscience and Remote Sensing, vol. 61, pp. 1-15, Jan. 2023.

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